A precision measurement of the Higgs boson mass in the diphoton decay channel

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Dedication

This research is dedicated to anyone who has struggled with imposter syndrome. It very nearly got the best of me.

Abstract

A measurement of the mass of the Higgs boson has been performed in the diphoton decay channel. This analysis is based on 138 fb^{-1} of pp collision data collected in 2016-2018 with the CMS detector at the LHC at a center-of-mass energy of 13 TeV. The mass is measured to be $m_H = 125.574 \pm 0.088$ (stat.) $\pm$ 0.121 (syst.) GeV. This thesis summarizes the strategy of this measurement with emphasis on the methods employed to mitigate and quantify the main sources of systematic uncertainty.

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Chapter 1

Introduction

The standard model (SM) of particle physics describes elementary particles and the forces by which they interact [?, ?, ?]. The SM is a phenomenological model built from over 100 years of experimental observation. It unifies the electromagnetic and weak forces and, along with the strong force, provides a coherent framework for the fundamental forces. Many successful predictions have been made using the SM, especially with respect to properties of particles which have been verified with unprecedented precision. One of these successful predictions is the Higgs mechanism and, by extension, the Higgs boson, which breaks the symmetry of electroweak interaction and explains how many particles obtain their mass [?, ?, ?, ?, ?, ?]. The Higgs boson was observed experimentally by the Compact Muon Solenoid (CMS) and A Toroidal LHC Apparatus (ATLAS) collaborations in 2012 using data collected at the European Center for Nuclear Research (CERN) Large Hadron Collider (LHC) [?, ?]. With the discovery of the Higgs boson, the particle content of the modern SM was completed.

While the SM does well to predict phenomena within its scope, there are many it does not describe at all. For example: the SM does not provide a description of the gravitational force, nor does it predict the existence of dark matter[1] or dark energy[2] [?]. And even within its scope, the SM has incomplete descriptions of particles such as neutrinos. These particles are

[1] A form of weakly interacting matter hypothesized as an explanation for some important phenomena in astrophysics

[2] The hypothesized energy source responsible for the expansion of space-time.

predicted to be massless, but experiments have confirmed that they are massive particles [**?**]. The SM also fails to address deeper problems in particle physics like the observed imbalance between matter and anti-matter in the universe [**?**] and the fine-tuning of the Higgs field vacuum expectation value [**?**, **?**]. Finally, the SM is not an elegant or concise model; it has many free parameters that must be provided by experiment [**?**]. And yet, it is exactly this abundance of free parameters that this thesis targets. In fact, this shortcoming will be leveraged against itself. Making precision measurements of the SM's free parameters is a way to not only test its self-consistency but also to shed light on areas of the SM where increased precision magnifies differences between prediction and measurement. Any differences in prediction and measurement open the door to physics beyond the SM (BSM physics) and offer potential solutions to many of the problems listed above.

Precision measurements of Higgs boson properties provide deeper insight into the theory of electroweak symmetry breaking[3], the nature of fermionic Yukawa interactions, and the shape of the Higgs field potential [**?**, **?**].

This thesis focuses on a precision measurement of the Higgs boson mass in the diphoton decay channel ($H\rightarrow \gamma\gamma$) using proton-proton collision data collected by the CMS experiment during Run 2 of the LHC. This thesis begins in Chapter **??** with a summary of the theoretical foundations of the SM that describe the Higgs boson and its relation to other particles, as well as summarizing the status of precision measurements of Higgs boson properties. Then, in Chapter **??**, the experimental apparatus, the CMS detector, is presented in detail and Chapter **??** describes how data collected by the CMS detector is processed into physics objects like electrons and photons to be used in analyses. Chapter **??** presents an improved method for correcting the energies of electrons and photons that allows analyses to make more precise measurements of particle properties. After this final set of corrections, Chapter **??** describes how $H\rightarrow \gamma\gamma$ events are grouped to improve the sensitivity of the Higgs boson mass measurement. Before discussion of the measurement itself, the statistical framework used to interpret results is laid out in Chapter **??**. Finally, the results of the

[3]This is the mathematical framework used to describe the origins of the Higgs boson, as well as its interactions with the other gauge bosons.

Higgs boson mass measurement are presented in Chapter **??** and a conclusion is given in Chapter **??**.

Chapter 2

The Standard Model and the Higgs boson

2.1 The Standard Model

2.1.1 Introduction

This chapter starts with an introduction to the components of the standard model and then delves into the mathematical formulation of these components and their interactions using a Lagrangian formalism. This formulation includes an introduction to gauge fields. Then, it provides descriptions of the electromagnetic and strong forces before covering the unification of the electromagnetic and weak forces into the electroweak force. This chapter also describes spontaneous symmetry breaking and how the Brout-Englert-Higgs (BEH) mechanism gives rise to the Higgs field and Higgs boson. Finally, this chapter concludes with a brief explanation of the production and decay modes of the Higgs boson at the LHC and a summary of the state of Higgs boson measurements to date.

2.1.2 Components of the Standard Model

Although the common vernacular is that the SM describes particles, the SM is, at its core, a field theory, so its components are fields. The particles observed in experiments are

excitations, and sometimes combinations of excitations, of the fields. However, for this description, they will be referred to as particles. The particles in the SM are separated primarily by their spin. Particles with integer spin values (0, 1) are bosons, and particles with half-integer spin values are fermions (-1/2, 1/2). Figure **??** shows a standard model particle diagram and will be a good reference for the following discussion.

The fermions in the SM are further divided by whether they interact via the strong force or not. Fermions interacting via the strong force and carrying color charge are called *quarks*. The quarks come in two types: up-type and down-type. The up-type quarks (up, charm, top) have an electric charge of $\frac{2}{3}e$, where e is the elementary unit of electric charge, and the down-type quarks (down, strange, bottom) have an electric charge of $-\frac{1}{3}e$. Quarks and other fermions have a distinguishing property called *generation*. The generation of a quark indicates its mass relative to other quarks, so the mass of the up quark is lighter than the charm mass, the charm mass is lighter than the top mass, and similarly for the down-type quarks. The remaining fermions that do not interact via the strong force are called *leptons*. Leptons interact through the electromagnetic and weak forces, have a charge of $-e$ in the case of the electron, the muon, and the tau, or carry no electric charge in the case of the three flavors of neutrinos. The SM also contains descriptions of the anti-particles of those listed above. However, the only observable difference between a particle and its anti-particle partner is the sign of its electric charge, so the SM's formulation includes them implicitly.

The SM also contains bosons, of which there are two types: vector bosons (spin 1) and a scalar boson (spin 0). The vector bosons include the mediators for the electromagnetic, weak, and strong forces. These bosons are the photon, the $W^{\pm}$ bosons, the Z boson, and the gluons, of which there are 8, respectively. Both the photon and the gluon are massless, but the $W^{\pm}$ and the Z bosons have masses around 80.4 GeV and 90.2 GeV, respectively [**?**]. Before discussing the final member of the SM, it is essential to note that the weak force, mediated by the $W^{\pm}$ and Z bosons, and the electromagnetic force, mediated by the photon, are one force: the electroweak force. The unification of these forces is discussed in detail below.

The final member of the SM is the scalar (spin 0) Higgs boson, whose existence comes about through spontaneous symmetry breaking in the electroweak sector. Coupling with the Higgs boson is the mechanism that gives mass to fermions and bosons in the SM.

2.1.3 Gauge invariant fields

Before diving into the details of gauge fields and the mathematical formulation of the SM it is worth taking a slight detour to talk about symmetries. Nöther's theorem [**?**] says that a system exhibiting symmetry will also have a conserved quantity. This theorem is invertible and can instead say that a system with conserved quantities must have an underlying symmetry. For example, the concepts of energy and momentum conservation apply to systems with underlying symmetries. In the case of conserved energy, the system is invariant under translations in time, and in the case of conserved momentum, the system is invariant under translations in space. This theorem is crucial to the understanding of the formulation of the SM.

The SM is a quantum field theory (QFT) formulated using a Lagrangian density [**?**]. The Euler-Lagrange equations and the principle of least action describe the dynamics of the SM and the allowed interactions [**?**]. The SM is a kind of *gauge field theory*, meaning that SM fields are invariant under *local gauge transformations*. Local gauge transformations are additions or alterations to a field's Lagrangian that shift the phase of the field. This phase shift in the field is allowed to be coordinate-dependent. Notably, the observable quantities derived from the SM fields are phase-independent, such as decay rates of massive particles and cross-sections of particle interactions. This requirement of invariance under gauge transformations is a type of symmetry and leads to two crucial features of the SM. The first is the use of gauge boson fields to describe the exchange of force between particles, and the second, according to Nöther's theorem, is the conservation of charge in the SM. Symmetries play an essential role in the SM, and the following sections describe the suite of symmetries observed in the SM, the interactions of the gauge boson fields with the fermion

fields, and how spontaneously breaking one of the following symmetries gives rise to the Higgs boson.

2.1.4 Local U(1) Symmetry: Quantum electrodynamics

The dynamics of the electromagnetic force in the SM, also referred to as quantum electrodynamics (QED), is a gauge field theory with a local U(1) symmetry, meaning that the QED Lagrangian $\mathcal{L}_{QED}$ is invariant under local U(1) transformations. The electromagnetic force, mediated by the photon field, interacts with the fermions. To demonstrate the invariance of $\mathcal{L}_{QED}$ under local U(1) transformations consider Dirac Lagrangian [**?**, **?**]:

$$\mathcal{L}_{\text{dirac}} = i\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi - m\bar{\psi}\psi, \tag{2.1}$$

where ψ is a Dirac spinor, $\bar{\psi}$ is its adjoint, γ^{μ} are the four Dirac matrices (4x4 matrices which obey the anti-commutation relations $\gamma^{\mu}, \gamma^{\nu} = 2g^{\mu\nu}$, where $g^{\mu\nu}$ is the Minkowski metric), and m is the mass of the fermion. Notice that the Dirac Lagrangian is not local U(1) invariant. By introducing a local U(1) transformation $e^{iq\theta(x)}$, where q and θ are real numbers and x is a spacetime coordinate, the Dirac spinors transform as:

$$\psi \xrightarrow{U(1)} e^{iq\theta(x)}\psi. \tag{2.2}$$

Because of the partial derivative in $\mathcal{L}_{\text{dirac}}$ there are additional terms remaining after the transformation:

$$\mathcal{L}'_{\text{dirac}} = \mathcal{L}_{\text{dirac}} - q\bar{\psi}\gamma^{\mu}(\partial_{\mu}\theta(x))\psi. \tag{2.3}$$

To preserve the symmetry of the QED Lagrangian, a new field A^{μ} must be added, which transforms as:

$$A^{\mu} \xrightarrow{U(1)} A^{\mu} + \frac{1}{g}\partial^{\mu}\theta(x). \tag{2.4}$$

This field becomes incorporated into the covariant derivative $D^{\mu} = \partial^{\mu} - iqA^{\mu}$ and the Lagrangian becomes

$$\mathcal{L} = i\bar{\psi}\gamma^{\mu}D_{\mu}\psi - m\bar{\psi}\psi. \tag{2.5}$$

This form of the QED Lagrangian is gauge invariant and describes the interaction of a fermion with a gauge boson A_μ, but does not describe the gauge boson itself. The gauge boson is a sustained excitation in a gauge field, and the field strength tensor describes the gauge field:

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu. \tag{2.6}$$

The kinetic term of the gauge field $-\frac{1}{4}F^{\mu\nu}F_{\mu\nu}$ adds to the QED Lagrangian. Its final form is:

$$\mathcal{L}_{QED} = i\bar{\psi}\gamma^\mu D_\mu \psi - m\bar{\psi}\psi - \frac{1}{4}F^{\mu\nu}F_{\mu\nu}. \tag{2.7}$$

Notice that there is no mass term associated with the photon field ($m_A A^\mu A_\mu$). The local gauge invariance requirement enforced in the Lagrangian forbids a massive gauge boson. And, since this is QED, the conserved charge associated with the local gauge symmetry is the basic electric charge ($q = e$ in Equation **??**). This demonstration is particularly relevant to building up to the weak and strong interactions because the number of generated bosons is equal to the number of degrees of freedom, or dimension, of the symmetry group [**?**]. The following sections describe the symmetry groups for the strong (SU(3)) and weak (SU(2)) interactions and the bosons generated by their symmetries.

2.1.5 SU(N) Symmetries: strong and weak interactions

The symmetry of the theory of QED is a local U(1) symmetry, but generally, gauge theories in physics have symmetries described by special unitary groups with dimension N (SU(N)). Recall from QED that the covariant derivative is $D^\mu = \partial^\mu + igA^\mu$, which preserves gauge invariance in the Lagrangian. In SU(N) symmetries, preserving gauge invariance requires the covariant derivative to take the form

$$D^\mu = I\partial^\mu + igT^a A^a_\mu \tag{2.8}$$

where I is the identity matrix of rank N, A^a_μ is the boson designated by a, g is the coupling constant (analogous to q from the U(1) symmetry), and T^a is the generator of the Lie group algebra. The generators T^a form a group with dimension $N^2 - 1$, and give rise to $N^2 - 1$

gauge bosons. The generator-to-gauge-boson feature extends from the U(1) theory of QED to SU(N) theories. For QED, the rank is 1, so the identity matrix is a scalar, the coupling strength q remained unchanged, the U(1) generator is the unity matrix (T = I), and the theory produced one gauge boson. However, SU(N) symmetries differ in complexity because they are non-abelian. The generators of non-abelian symmetries do not commute. Instead, they follow the commutation relation

$$[T^a, T^b] = if^{abc}T^c, \tag{2.9}$$

where f^{abc} are the structure constants of the group. QED's U(1) symmetry is abelian, so the structure constants are zero. The QED field strength tensor, associated with the field A^μ, took a simpler form because of the nature of the U(1) symmetry. In general, the field strength tensor in SU(N) is derived from commuting the covariant derivative with itself, using the relation

$$[D_\mu, D_\nu] = igT^a F^a_{\mu\nu} \tag{2.10}$$

and the field strength tensor associated with generator T^a is

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + gf^{abc}A^b_\mu A^c_\nu. \tag{2.11}$$

Notice that the field strength now depends on the term $gf^{abc}A^b_\mu A^c_\nu$. This term is a self-interaction term, i.e., a two-point interaction term. The expanded permutations of a, b, and c describe three and four-point self-interactions.

2.1.6 Strong interactions

The strong sector of the SM is described by an SU(3) gauge field theory and is known as quantum chromodynamics (QCD). Having set up the basis for SU(N), it is apparent that this group has eight symmetries. Traceless 3x3 unitary matrices represent the QCD generators, and the covariant derivative for this field is

$$D_\mu = \partial_\mu + ig_s\frac{\lambda^a}{2}A^a_\mu, \tag{2.12}$$

where the generators (λ^a) are the Gell-Mann matrices, g_s is the QCD gauge coupling, and A^a_μ are the eight gauge fields corresponding to gluons. These bosons are massless due to local gauge invariance, as we saw in QED, and carry color charge, which is analogous to electric charge, except that there are three independent charges labeled red, green, and blue. Furthermore, the only fermions that interact with gluons are the quarks since they are the only other SM particles to carry a color charge. This restricted set of interactions indicates that quarks transform as a triplet under color space transformations.

The form of the QCD Lagrangian in the SM is

$$\mathcal{L}_{QCD} = \sum_f i\overline{\psi}_f(\gamma^\mu D_\mu - m_f)\psi_f - \frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu}, \tag{2.13}$$

where the index f represents the six quark flavors, and m_f represents the mass of a quark with flavor f. The magnitude of the strong force exhibits a behavior known as asymptotic freedom, in which the force grows stronger with increasing distance instead of weakening, as is the case with QED. Asymptotic freedom leads to color confinement, a feature of QCD that prevents the existence of free particles with color charge. Color confinement is why quarks and gluons show up in particle detectors as showers of hadronic particles, known as jets.

2.1.7 Electroweak interactions

The unification of the electromagnetic and weak forces is a key component of the SM. Glashow, Weinberg, and Salam developed this model, known as the GWS model, in the late 1960s, starting the modern formulation of the SM [**?**]. The derivation of the GWS model begins with an SU(2) gauge field theory with three bosons. The covariant derivative is

$$D_\mu = \partial_\mu + i\frac{g}{2}W^{(i)}_\mu\sigma^{(i)} \tag{2.14}$$

where $W^{(I)}_\mu$ are the three SU(2) gauge fields, and the generators ($\sigma^{(i)}$) are the 2 x 2 Pauli spin matrices. The charge conserved in the weak sector charge is known as weak isospin. Both quarks and leptons have weak isospin charge and interact via the weak force. Quarks and leptons are doublets under weak gauge transformations. The weak interaction also

violates parity, meaning it is not invariant under spatial coordinate inversions. The weak force parity operators explicitly include this lack of invariance:

$$P_{L,R} = \frac{1}{2}(1 \mp \gamma^5) \tag{2.15}$$

where $P_{L,R}$ are the left and right-handed parity operators, and $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ anti-commutes with γ^μ making objects such as $\overline{\psi}\gamma^\mu\psi$ odd under parity transformations. The W bosons only couple to left-handed fermions, enabling this parity violation. While the form of Eq. **??** implies that this parity violation should hold for both the W and Z bosons, it is well known that Z bosons couple to both left and right-handed fermions. Treating weak isospin as the weak charge is a key insight into the GWS model and the unification of the electromagnetic and weak forces. By constructing the raising and lowering operators for this charge from the Pauli spin matrices and the currents that represent the exchange of the W bosons, it becomes apparent that the physical W bosons are linear combinations

$$W^\pm = \frac{1}{\sqrt{2}}(W_\mu^{(1)} \mp iW_\mu^{(2)}), \tag{2.16}$$

where $W_\mu^{(i)}$ are the SU(2) fields. Recall that the W bosons are linear combinations of other fields and that the Z boson couples to both left and right-handed fermions. From here, it is only a short jump to understand the electroweak force as a combination of two groups $\mathrm{SU}(2)_L$ x U(1). The charges of this combined force must be isospin and weak hypercharge Y. The covariant derivative incorporates the four fields for this model ($W_\mu^{(i)}$ and B_μ):

$$D_\mu = \partial_\mu + i\frac{g}{2}W_\mu^{(i)}\sigma^{(i)} + g'YB_\mu, \tag{2.17}$$

where g and g' are real numbers. The physical representation of the charged bosons is the same as in Eq. **??**, while rotations in the $\mathrm{SU}(2)_L$ x U(1) represent the neutral bosons as follows:

$$Z_\mu = \cos(\theta_W)W_\mu^{(3)} - \sin(\theta_W)B_\mu, \tag{2.18}$$

$$A_\mu = \sin(\theta_W)W_\mu^{(3)} - \cos(\theta_W)B_\mu, \tag{2.19}$$

where θ_W is the Weinberg angle. The left-handed fermions that these bosons couple to are $\mathrm{SU}(2)_L$ doublets, ψ_L, and the right-handed fermions are singlets, ψ_R. The right-handed parts of the electroweak Lagrangian are

$$\mathcal{L}_{EW} = -\overline{\psi}_R g' \cos(\theta_W) Y_\psi \gamma^\mu A_\mu \psi_R. \tag{2.20}$$

By comparing this to Eq. **??**, it is apparent that the coupling of the electron is $|e| = g' \cos(\theta_W)$, and the hypercharge Y_ψ is the particle's electric charge. The left-handed parts of the Lagrangian are

$$\mathcal{L} = -\overline{\psi}_L (g \sin(\theta_W) \frac{\sigma^{(3)}}{2} + g' \cos(\theta_W) Y_\psi) \gamma^\mu A_\mu \psi_L. \tag{2.21}$$

Similarly, the coupling here is $e = g \sin(\theta_W)$, and the hypercharges of the quarks and leptons are $\frac{1}{6}$ and $-\frac{1}{2}$ respectively. This theory violates parity in the weak sector and describes a massless boson, the photon, which couples identically to both left and right-handed fields.

The description of the interactions of SM fermions and bosons is now complete. However, it is worth noting that the previous section did not explain the origin of the mass of the particles described. As shown in Section **??**, the mass terms for bosons vary under local gauge transformations. While this variation does not affect the photon and the gluon because they are massless, it does affect the massive W and Z bosons. Furthermore, writing a simple Lagrangian for the mass of the fermions as

$$\mathcal{L} = -m(\overline{\psi}_L \psi_R + \overline{\psi}_R \psi_L) \tag{2.22}$$

cannot preserve gauge invariance due to the difference in transformations of left and right-handed spinors. Hence, the Lagrangian cannot include a mass term for the quarks and leptons. Resolving these issues requires a spontaneous symmetry breaking of the EW force, also known as the Higgs mechanism.

2.2 Spontaneous symmetry breaking in the electroweak sector

The Higgs mechanism generates the mass of the W and Z bosons through a spontaneous symmetry breaking in the electroweak sector. Peter Higgs, Robert Brout, and Francois Englert developed the Higgs mechanism in the 1960s [?, ?, ?]. This mechanism is best illustrated with a complex field ϕ and a locally gauge invariant Lagrangian, with fourth-order potential:

$$\mathcal{L} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + (D^{\mu}\phi)^{*}(D_{\mu}\phi) - m^{2}\phi^{*}\phi - \lambda(\phi^{*}\phi)^{2} \tag{2.23}$$

where λ is a positive real number, and the covariant derivative is the same as in Eq. **??**. If $m^2 > 0$ the potential has a minimum at $\phi_1 = \phi_2 = 0$, but if $m^2 < 0$ then the potential has degenerate minima at

$$|\phi| = \frac{-m^2}{2\lambda} \equiv \frac{v}{\sqrt{2}}, \tag{2.24}$$

where v is the vacuum expectation value (vev). The potential $|\phi|$ is symmetric and has degenerate minima, but the physical vacuum state cannot have infinitely many different values; the expectation value of ϕ ($< \phi >$) cannot be zero. Therefore, the symmetry is broken, hence the term spontaneous symmetry breaking. Figure **??** illustrates the symmetric potential and single-valued vev.

The direction of the ground state in the real-imaginary plane is arbitrary, so for demonstrative purposes, let it be in the real direction. Consider an expansion about the minimum with a field ϕ of the form

$$\phi = \frac{1}{\sqrt{2}}(v + \phi_1 + i\phi_2) \tag{2.25}$$

where ϕ_1 and ϕ_2 are real scalar fields. Inserting this definition into the Lagrangian in Eq. **??** produces mass terms of the form

$$\mathcal{L} = -2\lambda v^2\phi_1^2 + 2g^2v^2A^{\mu}A_{\mu}, \tag{2.26}$$

which shows a massive gauge boson A_μ. Hence, spontaneous symmetry breaking of a Lagrangian invariant under local gauge transformations produces a massive gauge boson. This

process generates the masses of the SM gauge bosons. Spontaneous symmetry breaking also creates a massive scalar boson ϕ_1, the Higgs boson.

The scalar Higgs field transforms as an SU(2) doublet H in the SM and has four degrees of freedom. The Higgs potential is quartic, and as such, the parts of the EW Lagrangian relevant to this potential are

$$\mathcal{L}_{EW} = (D^\mu H)^\dagger (D_\mu H) + \mu^2 H^\dagger H - \lambda (H^\dagger H)^2, \tag{2.27}$$

where μ and λ are positive real constants, and the covariant derivative comes from Equation **??**. Again, we see that this potential has degenerate non-zero minima at

$$H^\dagger H = \frac{\mu^2}{2\lambda} \equiv \frac{v^2}{2} \tag{2.28}$$

and without loss of generality, an expansion about the neutral component of the doublet's minimum reveals

$$H = \begin{pmatrix} 0 \\ \frac{v}{\sqrt{2}} + h(x) \end{pmatrix} \tag{2.29}$$

where $h(x)$ is the physical scalar boson of the Higgs mechanism. Expanding about the neutral component's minimum corresponds to choosing the unitary gauge. Applying the covariant derivative to the Higgs field and rotating using Equations **??** and **??** yields the Lagrangian terms

$$\mathcal{L}_{EW} = \frac{g^2 v^2}{4} W_\mu^+ W^{-\mu} + \frac{(g^2 + g'^2) v^2}{8} Z_\mu Z^\mu. \tag{2.30}$$

Hence, the W and Z bosons acquire mass while the photon remains massless. The Higgs mechanism explains the observed mass structure of the gauge bosons. Additionally, this theory predicts, and experiments have confirmed, the ratio of the W and Z bosons to be $\cos(\theta_W)$. Measurements of the observed SM bosons and fermions determine the value of the vev v.

The Higgs mechanism also adds mass terms for the fermions to the EW Lagrangian. These terms, known as Yukawa terms [**?**], take the form

$$\mathcal{L}_{EW} = g_f \overline{\psi}_L H \psi_R + h.c., \tag{2.31}$$

where g_f is the strength of the Yukawa coupling of the fermion to the Higgs boson and *h.c.* indicates the Hermitian conjugate. This term is now gauge invariant. Expanding the field H in terms of the vacuum expectation value and the scalar field $h(x)$, the mass terms for the fermions are

$$m_f = \frac{g_f v}{\sqrt{2}}, \tag{2.32}$$

The SM does not predict the strength of these couplings, g_f; experiments must measure their values.

The Higgs boson also interacts with the gauge bosons, and these three and four-point interactions appear in the EW Lagrangian as

$$\mathcal{L}_{EW} = m_W^2 \left(\frac{2h}{v} + \frac{h^2}{v^2}\right) W_\mu^+ W^{-\mu} + \frac{m_Z^2}{2}\left(\frac{2h}{v} + \frac{h^2}{v^2}\right) Z_\mu Z^\mu. \tag{2.33}$$

As before, the three- and four-point interaction terms' coupling strength is proportional to the gauge boson mass and the Higgs boson mass. The Higgs boson couples to itself, since it is a massive particle, through self-interaction terms in the Lagrangian, which take the form

$$\mathcal{L}_{EW} = -\lambda v^2 h^2 - \lambda v h^3 - \frac{1}{4}\lambda h^4, \tag{2.34}$$

from which the mass of the Higgs boson is $m_H = \sqrt{2\lambda}v$. The SM also does not predict a value of the Higgs boson mass; experiments must measure the value and return to the SM to make further calculations. After measuring the mass, many properties of the Higgs boson become apparent, including production and decay modes. The following section discusses the phenomenology of the Higgs boson and the experimental limitations imposed on measuring its properties.

2.3 Properties of the Higgs boson

One of the main goals of the LHC was to discover the Higgs boson, if it existed, thereby completing the SM. In 2012, the ATLAS and CMS collaborations announced the discovery of a new particle resembling the Higgs boson [**?**, **?**]. After this discovery, the focus of many research teams turned toward precision measurements of the properties of this new particle.

These precision measurements allow the SM to undergo rigorous self-consistency tests, which might produce evidence of physics beyond the standard model (BSM) or constrain the allowed spaces of BSM models. This section summarizes the most important production and decay modes of the Higgs boson and details the current state of precision Higgs boson measurements at the LHC.

2.3.1 Higgs boson production at the LHC

The high-energy proton-proton collisions at the LHC can produce Higgs bosons at rates that depend on the center-of-mass energy ($\sqrt{s}$) of the collisions and the mass of the Higgs boson. Gluon fusion (ggH) is the most common production mode for $\sqrt{s} = 13$ TeV and a Higgs boson mass of 125 GeV and contains only the Higgs boson in the final state. The final state of the ggH production mode is unique from other production modes in that the only particle in the final state is a Higgs boson. Other production modes have additional objects in the final state, like vector boson fusion (VBF), where the final state contains two quarks and a Higgs boson. Additional objects in the final state sometimes improve the ability to discriminate between Higgs boson signal and background, i.e., physics, which has the same signature objects and similar kinematics but comes from an initial state without a Higgs boson. The other most common production modes at the LHC are the production of a Higgs boson in association with a vector boson (VH) and the production of a Higgs boson alongside two top quarks (ttH). Table **??** lists the cross-sections for these production modes, and Figure **??** shows the leading order Feynman diagrams.

2.3.2 Higgs boson decay modes

The SM predicts the lifetime of the Higgs boson to be on the order of 10^{-22} s [**?**]. Such a short lifetime means that direct observation is impossible. Instead, experiments measure the decay constituents of the Higgs and infer its properties from these measurements. The relatively large mass of the Higgs boson allows it to decay through many channels or decay modes. Because the strength of any particle's coupling to the Higgs boson is proportional to

that particle's mass, the Higgs boson can decay directly to any massive particle, assuming proper conservation of other decay properties. The decay products must have a mass less than half that of the Higgs boson itself. However, a decay involving virtual particles of greater mass, which decay themselves, is allowed. Indirect decay, through loops, opens additional decay channels to massless particles, including gluons and photons. Table **??** lists the branching ratios of the seven principle decay modes. The $\gamma\gamma$, ZZ^*, $W^{\pm}W^{\mp *}$, and $\tau^+\tau^-$ decay modes were observed in Run 1 of the LHC [**?**, **?**, **?**]. The $b\bar{b}$ and $\tau^+\tau^-$ decay modes were observed in Run 2 of the LHC [**?**, **?**, **?**, **?**].

From Table **??**, it is apparent that the $b\bar{b}$ decay mode has the largest branching fraction. However, this channel is challenging to use in precision analyses due to the large hadronic background. The channels with the highest sensitivity are the diphoton ($\gamma\gamma$) and ZZ^* decays. The sensitivity of these channels relies on the small background and resolution of the reconstructed final states. As such, the diphoton and ZZ^* decay channels were the most vital contributors to the discovery of the Higgs boson and are the primary decay channels used in current precision analyses. The analysis presented in this thesis uses the diphoton decay channel. Figure **??** shows the three leading Feynman diagrams contributing to this process. The decay of the Higgs boson to a pair of massless particles can only happen through an intermediary loop, as shown, and the diphoton channel is explicitly sensitive to the Higgs boson's coupling to the top quark and the W boson.

2.3.3 Status of Higgs boson measurements

Many precision measurements of Higgs boson properties have been made since its discovery, including some tests of its structure. The first of these tests is the spin of the Higgs boson, which is the only fundamental scalar particle in the SM. Since it decays to two photons, the spin must be even. While the exact value has yet to be determined, dedicated measurements by ATLAS and CMS have found that the Higgs boson is not a spin-two particle nor a spin-zero particle with negative parity (a pseudoscalar) [**?**, **?**]. The possibility that the Higgs boson is a mixture of scalar and pseudoscalar states is an active area of research.

Another area of precision measurements is the coupling of the Higgs boson to other SM particles. These measurements compare the rate of Higgs boson production to the production rate predicted by the SM. The simplified template cross-section (STXS) framework is used for these measurements [**?**]. The STXS framework is a modern approach to Higgs boson precision measurements that reduces the dependence of the measurements on theory while still allowing the use of advanced analysis techniques. The latest results indicate a firm agreement with the SM with a total signal strength of 1.12 ± 0.09 relative to the SM prediction [**?**]. Figure **??** shows the Higgs boson signal strength measured in the primary production channels.

Mass is another area of focus in precision measurements of Higgs boson properties. This measurement uses the $\gamma\gamma$ and ZZ* decay modes, the only channels with sufficiently narrow mass resolution. During Run 1, the ATLAS and CMS experiments collaborated to produce a Higgs boson mass measurement of $m_H = 125.09 \pm 0.24$ GeV, which is a relative precision of 0.2% [**?**]. The precision of the Higgs mass measurement improved after measurements made by CMS in the $\gamma\gamma$ and ZZ* channels using Run 1 data and the 2016 dataset. The Higgs boson mass was measured to be $m_{\mathrm{H}} = 125.38 \pm 0.14$ GeV; this measurement has a relative precision of 0.11% [**?**]. Figure **??** summarizes the current Higgs mass measurements made by the CMS collaboration.

2.4 Summary and strategy

The SM is a gauge field theory that describes the fundamental particles and the forces that govern their interactions, excluding gravity. Predictions made by the SM have been tested at unprecedented precision, and the final component of the SM, the Higgs boson, was observed in 2012. The Higgs boson is an essential piece of the SM as it explains the origins of mass for both the gauge bosons and the fermions, except neutrinos. Measuring the mass of the Higgs boson completes the model of the Higgs mechanism since the strength of the coupling to the Higgs field depends on the mass of the Higgs boson. Hence, precisely measuring the properties of the Higgs boson is imperative in advancing current research

in particle physics. This analysis aims to measure the mass of the Higgs boson in the diphoton decay channel using the Run 2 dataset collected by the CMS experiment at the LHC, thereby constraining SM predictions to test its self-consistency. The strategy for this precision measurement uses the excellent resolution of the CMS experiment's electromagnetic calorimeter to overcome the small branching fraction of the diphoton decay channel and its irreducible continuum background. In addition, modern analysis techniques, such as multivariate classifiers, are used to further distinguish diphoton events from the background by predicting their production mode. Finally, the diphoton invariant mass distribution in signal and background are fit. The signal model is taken from the simulation, and the background model is obtained from a fit to sidebands in data.

Chapter 3

The CMS experiment

3.1 The Large Hadron Collider

The Large Hadron Collider (LHC) is a proton-proton collider sited near Geneva, Switzerland, in a 27 km long underground tunnel that holds a vacuum pipe containing two counter-rotating beams of protons. These protons are steered using magnets and collide at four interaction points (IPs). Two IPs are central to general-purpose detectors: the Compact Muon Solenoid (CMS) and A Toroidal LHC Apparatus (ATLAS). Figure **??** shows the location of these detectors and the layout of the LHC. Two of the main goals of the LHC program are to determine the mechanism of electroweak symmetry breaking and to search for physics beyond the SM. As such, the LHC was designed to implement proton-proton collisions with a maximum center of mass energy of 14 TeV and a maximum luminosity of $10^{34}\text{cm}^{-2}\text{s}^{-1}$.

The construction of the LHC was a massive undertaking, and several past experiments and their facilities were used to conserve space and funding. For example, the LHC's tunnel previously held the CERN's large electron-positron collider, and CERN's proton synchrotron (PS) and super proton synchrotron (SPS) are now injectors to the LHC.

The process of preparing the beams is complex and contains multiple steps. Initially, protons are extracted from a hydrogen plasma and directed through linear accelerators and

proton synchrotrons. The acceleration sequence proceeds as follows: it begins with the acceleration of protons extracted from a hydrogen gas plasma to an energy of 50 MeV using a series of linear accelerators. From there, they are injected into the Proton Synchrotron (PS) booster ring, where they are bunched and accelerated to an energy of 25 GeV. Next, the protons are transferred to the Super Proton Synchrotron (SPS) booster, where they reach an energy of 450 GeV. Finally, the proton bunches are injected into the two rings of the LHC, where they reach an energy of 7 TeV.

The beams in the LHC are steered using 1232 superconducting dipole magnets cooled to 1.9 K, which produce magnetic fields up to 8.4 T. RF cavities are used to accelerate the beams and to compensate for the energy loss due to synchrotron radiation once the beams have reached their full energy. At the end of the cycle, the LHC contains up to 2808 bunches of protons separated by 7.5 m that collide in the IPs at 25 ns intervals, corresponding to a bunch collision rate of 40 MHz. Due to the large number of protons in each bunch, there are, on average, 35 separate proton-proton collisions in each bunch crossing.

For the three years that the CMS experiment collected data for this analysis, the LHC operated at a center of mass energy of 13 TeV and a bunch spacing of 25 ns. The delivered luminosity in 2016, 2017, and 2018 were 41.58 fb^{-1}, 49.79 fb^{-1}, and 67.86 fb^{-1}, respectively. The data collected during this three-year period is known as Run 2. Figure **??** shows the cumulative integrated luminosity of Run 2.

3.2 Overview of the CMS detector

The CMS detector is located at Point 5 in the LHC ring. It is a general-purpose detector whose central feature is a high-magnetic field solenoid. Figure **??** shows the main features of the detection, and Figure **??** shows the functionality of the parts of the detector, known as sub-detectors. The superconducting solenoid from which the CMS detector gets its name has a radius of 6.3 m, a length of 12.5 m, and produces a magnetic field of 3.8 T. The pixel and strip silicon trackers, the lead-tungstate ($PbWO_4$) crystal electromagnetic calorimeter (ECAL), and a brass and plastic scintillator hadron calorimeter (HCAL) are all located

inside the solenoid. A steel return yoke outside the solenoid returns the magnetic field flux. Embedded in the return yoke are gas-ionization chambers used for detecting muons. For further information on the CMS detector, see Ref. [**?**]. A description of the components of the CMS detector used to measure the decay of the Higgs boson to two photons, the subject of this thesis, follows.

3.2.1 Detector coordinate system

An essential part of any measurement is establishing a standard Cartesian coordinate system used through the detector. In CMS, the detector uses a right-handed coordinate system. The origin of the system is at the IP. The x-axis points toward the center of the LHC ring; the y-axis points upward; and the z-axis points in the direction of the counter-clockwise beam as viewed from above. In addition to the rectilinear coordinate system, a cylindrical coordinate system is used. In this system, the radial distance r is defined in the x-y plane, the azimuthal angle ϕ is measured from the positive x-axis, and the polar angle θ is measured from the positive z-axis.

It is often helpful to express the production angle in terms of the particle's rapidity, defined as

$$y = \tanh^{-1}\left(\frac{p_z c}{E}\right) \tag{3.1}$$

where p_z is the momentum of the particle along the z-axis, c is the speed of light, and E is the energy of the particle. Differences in rapidity are Lorentz invariant to boosts along the z-axis, making it useful for determining angles between particles. For highly relativistic particles whose mass is negligible, rapidity reduces to pseudorapidity (η), defined as:

$$y \approx \eta = -\ln\tan\left(\frac{\theta}{2}\right). \tag{3.2}$$

Rapidity and pseudorapidity range from ($-\infty$,∞) for $\theta = (\pi, 0)$, and is 0 when $\theta = \frac{\pi}{2}$ in the central plane of the detector. The CMS detector can measure particles with pseudorapidities between -5 and 5, with full coverage of all the sub-detectors ranging from

-2.5 to 2.5 in pseudorapidity. Other commonly used variables include a particle's transverse momentum $\overline{p}_T$ and transverse energy E_T. These represent the magnitude of a particle's momentum and energy perpendicular to the beamline, and assuming the particle to have negligible mass, are defined as:

$$|\overline{P}_T| \approx E_T \approx \frac{E}{\cosh \eta}. \tag{3.3}$$

3.3 The Tracker

From the IP, the tracker is, radially, the first sub-detector that particles encounter. Charged particles interact with the various parts of the tracker, and the helical tracks measured from these interactions are collected and fit to determine their momenta. The tracks also help determine the location of the interaction's vertex or point of origin. When the LHC is in operation, each bunch crossing produces ~35 interactions with ~1000 charged particle tracks.

The tracker's crucial design elements included the detector's granularity, measurement speed, and material radiation hardness. This sub-detector comprises silicon pixel and strip detectors and covers the region -2.5 to 2.5 in pseudorapidity.

3.3.1 The Pixel tracker

The silicon pixel tracker is the closest sub-detector to the IP, as shown in Figure **??**, and is wholly immersed in the 3.8T magnetic field from the solenoid. This highly granular layered cylindrical detector is composed of 66 million silicon pixels and produces a 3D track of every charged particle within its coverage of -2.5 to 2.5 in pseudorapidity. The silicon pixels are 100×150 μm^2 in r$\phi \times z$ and have a resolution of 10 μm along the r-axis and 20-40 μm along the z-axis. The pixel size was explicitly chosen to keep the pixel occupancy below 10^{-4} per pixel per bunch crossing at $r = 10$ cm. Additionally, the tracker takes advantage of the Lorentz effect from the magnetic field by using charge sharing among the pixels. To further exploit this feature, the pixel modules in the endcaps of the tracker overlap and are feathered at an angle of 20 degrees.

3.3.2 The Silicon Strip tracker

The strip tracker is the next layer of the tracker and surrounds the pixel tracker, as shown in Figure **??**. It is divided into three regions: the barrel and two endcaps. The barrel component of the strip tracker consists of two parts: the tracker inner barrel (TIB) and the tracker outer barrel (TOB). The TIB has four layers of silicon strips made with 320 μm thick silicon sensors, with 2724 sensors. The first two layers provide a "stereo" measurement in both $r \times \phi$ and $r \times z$ with a stereo angle of 100 mrad; as such, the resolution in the TIB ranges from 24 - 36 $\times$ 23 μm in $r\phi \times z$. The TOB comprises six layers of sensors with a thickness of 500 μm with a total of 5208 sensors. Again, the first two layers provide a "stereo" measurement with an angle of 100 mrad, which results in a resolution ranging from 35-53 μm $\times$ 52 μm in $r\phi \times z$.

The endcaps of the strip tracker have two components: the tracker endcap (TEC) and the tracker inner discs (TID). The silicon sensor modules are oriented perpendicular to the beamline to ensure hermetic coverage out to -2.5 and 2.5 in pseudorapidity. The TEC comprises nine layers of between four and seven rings of modules, while the TID covers the region between the TIB and the TEC and is composed of 3 layers of 3 rings of modules. The TID sensors and the first four layers of the TEC have a thickness of 320 μm and the remaining five layers of the TEC have a thickness of 500 μm. Layers 1, 2, and 5 of the TEC are "stereo" measurement modules with a stereo angle of 100 mrad.

3.3.3 Performance of the Tracker

The track reconstruction efficiency is measured using muons and was greater than 99% for Run 2, as shown on the left in Figure **??**.

The momentum resolution of the tracker, shown in the center in Figure **??**, was studied in simulation for the 2016 and 2017 CMS detector and ranged from 1% in the inner barrel to 5% in the forward regions. Additionally, the vertex resolution along the z-axis of simulated tracks ranges from 100 μm to 10 μm as the number of tracks increases.

3.4 The Electromagnetic calorimeter

The CMS ECAL is a highly granular, heterogeneous crystal calorimeter composed of 75,848 1.2kg lead-tungstate ($PbWO_4$) crystals. Its primary function is to measure the energy of electrons and photons produced in proton-proton collisions. Additionally, the triggers use the ECAL's measurements to select events of interest.

The ECAL has three primary components: the ECAL barrel (EB) covering η between -1.479 and 1.479 and the two ECAL endcaps (EE) that cover the regions $1.479 < |\eta| < 3.0$. In front of the endcap, covering the regions $1.653 < |\eta| < 3.0$, are the ECAL pre-showers (ES). The ES detector has two layers of silicon strip sensors between lead plates. Figure **??** shows the layout of the ECAL.

The EB has two halves: EB+ and EB-. Each half has 18 super-modules, containing four modules and 1700 crystals. Modules contain 40 to 50 lightweight glass-fiber boxes called sub-modules grouped in sets of 10. The crystals are mounted in a feathered pattern at an angle of 3 degrees with respect to the direction from the interaction point to prevent particles from passing through cracks between crystals.

The EE is also separated into EE+ and EE-, each containing 2 *dees*. Each *dee* holds 3662 crystals arranged in an x-y geometry. These crystals are grouped in sets of 5×5 to form super-crystals.

The ES sits in front of the endcaps and helps improve the spatial resolution in the endcap regions. The silicon strips in the ES are 2 mm wide, compared to the front face of the ECAL crystals at 3 cm. Photons and electrons interact in the layers of lead in the ES and initiate an electromagnetic shower, with between 6% and 8% of the energy of an EM shower deposited in the ES. When large energy deposits are seen in the ECAL endcaps, the spatial information of the ES is used to determine if it was a single high-energy photon or a pair of lower-energy photons.

3.4.1 Lead Tungstate ($PbWO_4$) Crystals

The main processes in electromagnetic showers produced by high-energy photons and electrons in the crystals are electron pair production and bremsstrahlung. This cascade process of electrons producing photons through bremsstrahlung, and photons pair producing into pairs of electrons, continues until the energy of the particles becomes too small to support pair production. At this point, the energy loss continues as ionization, Compton scattering, and the photoelectric effect. A scintillator is characterized by three properties: the radiation length χ_0, the Molière radius R_M, and the scintillation decay time τ_{sc}. The radiation length is the distance traveled in the material by an electron before losing 1/e of its initial energy. The Molière radius is defined as the radius of a cone that contains 90% of the energy of an EM shower in the material. The scintillation decay time defines the time it takes for the intensity of output scintillation light to drop to 1/e of its maximum. For the CMS ECAL, it is essential to have a compact detector, meaning a small radiation length and Molière radius. Because of the high event rate, the scintillation decay time needs to be very fast to prevent signals from consecutive events from overlapping.

The scintillation material of the ECAL was chosen to be lead tungstate ($PbWO_4$) due to its radiation hardness, its short radiation length ($\chi_0 = 0.89$ cm), its small Molière radius ($R_M = 2.2$ cm), and its fast scintillation decay time ($\tau_{sc} = 25$ ns), and its relatively low cost. These properties lead to excellent spatial resolution and separation because the electromagnetic showers are well contained, and the fast scintillation decay time ensures that showers from different bunches do not overlap in the detector [**?**].

3.4.2 Photodetectors

Lead tungstate has a lower light yield than other scintillators commonly used in particle detectors, producing 30 photons/MeV. Such a low light yield requires photodetector signals to have a high quantum efficiency and low noise.

The photodetectors used in the EB, avalanche photodiodes (APDs), are unaffected by the presence of the magnetic field. However, they are more sensitive to radiation damage than the vacuum phototriodes (VPTs) used in the EE, as discussed below.

Attached to the rear face of the crystals in the EB are two 5 mm $\times$ 5 mm APDs. These do not fully cover the rear face of the crystal and collect an average of 4.5 scintillation photons per MeV of energy deposited in the crystal. Their average gain is 50, and their operational voltage is $\sim$400 V. The ADPs will degrade due to the high radiation environment. Initially, the dark current in the APDs was a few nanoamperes. The dark current increases with radiation and will reach values of the order 5 nA at the end of LHC operation.

The expected total radiation dose of the EE is greater than the EB's. For an integrated luminosity of 500 fb^{-1}, or roughly ten years of LHC operation, the EB will receive a total dose of 3 kGy. In contrast, the EE will receive an expected 3 kGy, 20 kGy, and 200 kGy at values of η = 1.5, 2, and 3, respectively. The EE VPTs were chosen for their tolerance to radiation and because using APDs would produce a much higher leakage current under a similar radiation dose [?]. VPTs are custom photodetectors with a single gain and a copper mesh anode. In the VPTs, the electron drift is parallel to the magnetic field, allowing for high magnetic field operation. Irradiation and magnetic field testing showed a loss of less than 10% in the mean response at the expected radiation dose attributed to the degradation of the photocathode.

3.4.3 Laser Monitoring

While its radiation hardness is a primary feature that led to the selection of lead tungstate, it is not entirely unaffected by radiation. As the crystals are exposed to high-intensity radiation, they develop color centers in the crystal lattice that absorb scintillation photons as they propagate through the crystal, reducing the transparency. To monitor and correct for this loss of transparency, a system of lasers injects light into the crystals, and photodiodes measure the lasers' signals. They are fired during the abort gap, a gap in the LHC bunch train, to avoid overlap with beam-related events. The abort gap is an intentional gap in

the proton beams of 7 μs, allowing kicker magnets to fire and dump the beams in case of an emergency. The laser system measures the response of ~600 crystals during each abort gap, and the entire ECAL is measured every 40 minutes.

The measure of the crystals' transparency is the relative response, which is the ratio of the current response of the laser in a crystal to the initial response of the laser in a crystal when the detector was first installed. The relative response of the ECAL in several rings of $|\eta|$ is shown in Figure **??**. The response of the crystals changed by ~5% in a year of data taking during Run 2, while the response of crystals in the EE changed by up to 20% in a year. The crystals anneal at room temperature between runs, increasing their transparency and, hence, their relative response. The effect of the annealing can be seen in Figure **??** as the increase in the relative response of the crystals between data-taking periods. More information on the impacts of ECAL crystal irradiation is provided in Chapter **??**.

3.4.4 Calibration and Energy Resolution

The energy resolution of the ECAL was measured using a 3x3 arrangement of EB crystals in test beams, and the takes the form:

$$\frac{\sigma(E)}{E} = \frac{2.8\%}{\sqrt{E}} \oplus \frac{12\%}{E} \oplus 0.3\% \tag{3.4}$$

where E is the energy of a particle in GeV, and σ is the energy resolution. The three terms, in order, are the stochastic term, the noise term, and additional constants. The stochastic term accounts for statistical fluctuations in the EM showers in the crystals, including the formation of the shower and fluctuations in photodetector sampling. The noise term comes from the electronic noise in the pre-amplifier and the APDs. Finally, the constant term, which dominates at high energies, accounts for energy losses into non-scintillating material, energy lost to the EM showers leaking out of the crystals and into other sub-detectors, and errors in the energy response calibration [**?**].

Precisely measuring the energy of an electromagnetic shower with the ECAL requires careful calibration, often at the level of individual crystals. The EM showers that develop

in the ECAL deposit energy in many crystals, spreading in the ϕ direction due to the magnetic field's effect on charged particles. A clustering algorithm groups nearby crystals into structures called "superclusters" (SC). These SCs are used to measure the energy of an EM object using the following equation:

$$E_{e/\gamma} = F_{e/\gamma} E_{SC} = F_{e/\gamma}(G \sum_i S_i(t) C_i A_i + E_{ES}). \tag{3.5}$$

The elements of this equation are as follows:

1. $F_{e/\gamma}$: Corrections applied to the energy that mainly account for effects from upstream material and effects due to the geometry of the ECAL and imperfect clustering. These corrections are derived using a multivariate regression.

2. G: ADC-to-GeV conversion factor, which functions on a global scale. This global scale is derived by requiring that the measured mass of the Z boson matches its known value[?].

3. $S_i(t)$: Corrections for the loss of scintillation light to color centers formed as the ECAL crystals are irradiated. The relationship between the measured laser response and the scintillation corrections is $\frac{S(t)}{S_0} = \left(\frac{R(t)}{R_0}\right)^{\alpha}$ where $S(t)$ is the scintillation response of the crystal at time t, S_0 is the initial scintillation response of the crystal, $R(t)$ is the laser response of the crystal at time t, R_0 is the initial laser response of the crystal, and α is a material dependent quantity.

4. C_i: this is a calibration correction called the "intercalibration". It accounts for the differences in scintillation yield between crystals, which can be as large as 15% for crystals in the EB and as large as 25% for crystals in the EE. The intercalibrations are determined using the ϕ-symmetry of the ECAL and the known mass of π^0 and η mesons measured through their decay to two photons.

5. A_i: The amplitude of the response of each crystal i.

The calibrations listed above successfully calibrate the energy scale and improve the energy resolution of the EM showers measured by the ECAL. The final energy resolution on unconverted photons with high transverse momentum is approximately 1%. These are the photons primarily used in H$\rightarrow \gamma\gamma$ analyses.

3.5 Hadronic calorimeter

The hadron calorimeter (HCAL) subdetector measures the energies of hadronic particles, typically coming from jets. It comprises layers of brass, which acts as an absorber, and plastic scintillator tiles read out using wavelength-shifting fibers. Brass was chosen as the absorber material because it is non-magnetic, has a short interaction length, and is relatively easy to machine. This combination of materials maximizes the nuclear interaction length inside the magnet coil to better contain the hadronic showers and minimize non-Gaussian tails in the energy resolution [**?**]. The scintillator tiles are 3.7 mm thick, are sandwiched between the brass plates, and have full coverage in ϕ. A diagram of the HCAL and its geometry is shown in Figure **??**.

The HCAL barrel (HB) contains 15 layers of 5cm thick brass plates between 3.7 mm scintillator tiles and covers the pseudorapidity range $|\eta| < 1.4$. The layer of HB closest to the ECAL is a 9mm thick scintillator. The HCAL endcap (HE) covers the regions in pseudorapidity between $1.3 < |\eta| < 3.0$. The part of the HCAL that sits outside the solenoid (HO) covers the region $|\eta| < 1.26$ and is responsible for improving the energy resolution in the tails of hadronic showers and for improving the energy resolution of the missing transverse energy. The HCAL detector's final component lies in the detector's forward region (HF). It covers the highest ranges of pseudorapidity between $3.0 < |\eta| < 5.0$ and is made of steel and quartz fiber.

3.6 Muon detection systems

The muon detector is the last subdetector and the furthest from the interaction point. This subdetector makes precise measurements of muons' position, momentum, and charge. The muon detection system is composed of a barrel and endcap in much the same fashion as the other detectors and is embedded in the iron return yolk of the magnet. The layout of the muon system is shown in Figure **??**.

3.6.1 Drift tube chambers

The drift tube chambers (DT) measure the position of muons in the barrel portion of the detector. Each tube is 4cm wide and contains a stretched wire suspended in a gas of 85% argon and 15% CO_2. The DT chambers comprise cells bounded by aluminum plates and I-shaped aluminum beams, which serve as cathodes. In the center of the cell is a stretched stainless steel wire, which functions as an anode. The efficiency of DT chamber cells is $\sim$99.8%, and each cell has a spatial resolution of $\sim$180μm. Free electrons produced by ionizing particles drift to the anode wire, where gas amplification occurs. The time between the particle crossing the tube and the signal on the wire is used to measure the location of the ionizing particle's interaction with the DT.

3.6.2 Cathode strip chambers

The CMS muon detector has 468 cathode strip chambers (CSCs), which are trapezoidal in shape and cover either 10° or 20°, that detect the location of muons. The CSCs contain six anode wire plans set between seven cathode panels. The wires run azimuthally in the detector and are used to define a muon track's radial coordinate. The muon coordinate in ϕ is measured by interpolating the charge induced on the strips.

3.6.3 Resistive plate chambers

The muon detection system also contains gaseous parallel-plate detectors known as resistive plate chambers (RPCs) that tag the time of an ionizing event. The RPCs measure the t_0

used in the DT measurements of the location of the ionizing particle. Six layers of RPCs are embedded in the barrel iron return yoke, with some redundancy in coverage to ensure the trigger algorithm always has four layers of RPC information to use in reconstruction. This guarantees the reconstruction of low-energy ionizing particles, which may stop inside the return yolk.

3.7 Data acquisition

In the LHC, proton-proton collisions occur every 25 ns, corresponding to an event rate of 40 MHz. Many of these events contain interactions that are of little interest to CMS. A trigger system reduces the event rate and focuses on interesting physics. This system takes place in two steps or levels. The Level 1 (L1) trigger is entirely hardware based since it must be extremely fast, the second trigger are software based and is collectively referred to as the High Level Trigger (HLT), where ∼12,000 processors are used to select events for storage.

3.7.1 Level 1 trigger

The L1 trigger uses many FPGAs to reduce the event rate from 40 MHz to 100 kHz during the L1 decision time. Each event and all its information is stored in an on-detector buffer. To prevent mixing particle information from different events, the detectors must have a precise time alignment, and the signals from the millions of readout channels in the detector must be synchronized. The latency tolerance is about 4 μs. If the event passes the L1 trigger, it is processed by the HLT. To keep up with the event rate, the L1 trigger only uses information from a few subdetectors: the calorimeters, and the muon chambers. The architecture of the L1 trigger is shown in Figure **??**.

3.7.2 High level trigger

When the L1 trigger selects an event, data is read from the detector and sent to the HLT for processing. The HLT program runs on a farm of commercial processors, with one event per processor, and reduces the event rate from 100 kHz to 1 kHz. To pass the HLT an

event must satisfy one or more predefined event selection criteria. The criteria are known as paths and are organized by computational speed, with the least expensive paths checked first. An event may pass multiple paths, but it is stored as soon as it passes a single path to improve processing speed. After passing the HLT, an event is saved to disk for later processing by offline software, which produces high-level objects used in physics analyses, such as electrons, photons, and jets.

Chapter 4

Event reconstruction and selection

4.1 Introduction

This analysis aims to measure the mass of the Higgs boson as precisely as possible. For this, the diphoton invariant mass ($m_{\gamma\gamma}$) must be precisely reconstructed from the output of the CMS detector. The H$\rightarrow \gamma\gamma$ signal is narrowly peaked and sits on a smoothly falling background of diphoton events, primarily coming from $gg \rightarrow \gamma\gamma$. For events with two reconstructed photons, the invariant mass is defined to be

$$m_{\gamma\gamma} = \sqrt{2E_1^\gamma E_2^\gamma (1 - \cos(\theta_{1,2}))} \tag{4.1}$$

where E_1^γ and E_2^γ are the energies of the photons in the event, and $\theta_{1,2}$ is the opening angle between them. Several factors that determine the overall precision of the Higgs boson mass measurement are:

1. *Diphoton mass resolution ($\sigma_{\gamma\gamma}/m_{\gamma\gamma}$)*: the narrower the signal peak, the easier it is to isolate from the diphoton background. The dominant contributions to this aspect of the analysis come from precise energy calibration and event categorization. These topics are discussed in Chapters **??**, and **??** respectively.

2. *Systematic uncertainties on the diphoton invariant mass*: the more precisely the photon energies are known, the more precisely the diphoton invariant mass is known.

However, a significant caveat is the uncertainty on the opening angle $\theta_{1,2}$. If the primary vertex - the point of origin for the diphoton event - can be determined to within 1 cm, then this uncertainty becomes insignificant, and the uncertainty on the diphoton invariant mass is dominated by the uncertainties associated with the photon energies, otherwise it contributes significantly to the uncertainty. Details on systematic uncertainties affecting the photon energies are described in Chapter **??**.

This chapter focuses on reconstructing physics objects, such as photons and electrons, using the particle flow algorithm [**?**] and the vertex and photon identification process used in H$\rightarrow \gamma\gamma$ analyses.

4.2 Particle flow

Particle flow (PF) reconstruction [**?**] is an offline analysis technique that uses tracks or energy clusters from each CMS sub-detector to build physics objects. This process is referred to as "reconstruction." The CMS detector uniquely benefits from the PF approach due to its highly granular subdetectors and its high magnetic field. The high granularity leads to detailed clustering algorithms in the ECAL and the HCAL, while the magnetic field allows charged particles to be tracked as they bend. The PF reconstruction process uses the information from all subdetectors to determine the optimal classification of objects in the detector.

The reconstruction process begins at the interior of the detector and works outward. Interactions with the pixel tracker are grouped into tracks by an algorithm based on Kalman Filtering [**?**]. This process uses an iterative approach to reduce the computational complexity by removing tracks easily paired with energy deposits. The first iteration picks out tracks with at least three pixel hits; this makes up the majority of reconstructed tracks. The second iteration picks out tracks that have only two pixel hits. The third iteration finds prompt tracks with $p_T < 0.8$ GeV. Iterations four through six attempt to identify tracks originating outside the beam spot and recover any tracks not found in the previous iterations. The muon system's tracking procedure functions similarly.

The calorimeters (ECAL and HCAL) use clustering algorithms to group energy deposits from a particle shower. The process for the ECAL is described here due to its importance in the H$\rightarrow \gamma\gamma$ analysis; the HCAL uses a similar procedure. The clustering algorithm starts by selecting *seed crystals* that have an energy above a prescribed threshold and have the highest local energy deposition, i.e., the energy deposited in the seed crystal is higher than the energy of any surrounding crystals. From these seed crystals, topological clusters are built by adding neighboring (by edge or corner) energy deposits to the cluster. Crystals belonging to more than one topological cluster share the energy between the allowed clusters, assuming Gaussian shower distributions. Once the topological clusters are built, they are further grouped into superclusters (SCs). This additional grouping accounts for energy losses due to bremsstrahlung since the electromagnetic showers have similar η values but tend to spread in ϕ because of their interaction with the magnetic field.

The PF algorithm uses the tracks and energy clusters described above to build collections of candidates for five types of particles: muons, electrons, photons, neutral hadrons, and charged hadrons. The detector features used to construct these objects are described in Table **??**. The rest of this chapter describes the use of these objects in the H$\rightarrow \gamma\gamma$ analysis.

4.3 Samples

4.3.1 Data

The data used in this analysis comes from p-p collisions collected by the CMS experiment at $\sqrt{s} = 13$ TeV during the Run 2 data-taking period at the LHC. The CMS experiment collected data in 2016, 2017, and 2018 and measured in an integrated luminosity of 137 fb^{-1} with 35.9 fb^{-1}, 41.5 fb^{-1}, and 59.4 fb^{-1} collected in each year respectively.

Events in data are selected by the trigger system described in Section **??**. The trigger requirements are shown in Table **??**. The efficiency of the trigger selection is measured by a *tag-and-probe* technique that exploits the well-known properties of the Z boson decay. In this technique, the *tag* object in the Z boson decay is subject to stringent selection criteria. In contrast, the *probe* object is subject to less stringent criteria, and the invariant mass of

the tag-and-probe pair is required to be near that of the Z boson mass. For any selection criteria C, the efficiency of the selection criteria is the ratio of events passing C to the total number of events. This method of determining the selection efficiency requires that the selection criteria applied to the probe do not affect the efficiency of the selection criteria C.

It is common in H$\rightarrow \gamma\gamma$ analyses to use Z$\rightarrow ee$ decays to validate and measure many detector and analysis properties such as efficiencies because the ECAL reconstructs both electrons and photons, and the Z boson mass is reasonably close to that of the Higgs boson. The tag-and-probe method described above is applied to electrons whose η and R_9 distributions have been reweighted to match those distributions in H$\rightarrow \gamma\gamma$ events. The efficiency of the trigger is evaluated in bins of p_T, η, and R_9 for each SC. The variable R_9 is defined as the ratio of energy deposited in the 3×3 block of crystals surrounding the seed crystal, divided by the energy of the SC ($\mathrm{R}_9 = E_{3\times3}/E_{\mathrm{SC}}$). R_9 represents the degree to which a shower is columnated, with values near 1 indicating a highly columnated shower, and values near 0 indicating a very loosely columnated shower. Loosely columnated showers, *i.e.* lower values of R_9, are correlated with objects that have had an interaction prior to reaching the ECAL. The trigger efficiencies for SCs in the EB and EE are 97% and 95%, respectively, and the product of these per SC efficiencies is used to weight simulated diphoton events to match the trigger efficiency observed in data. However, the trigger selections are much looser than the offline selection criteria used in the H$\rightarrow \gamma\gamma$ analyses.

4.3.2 Simulation

H$\rightarrow \gamma\gamma$ analyses use Monte Carlo (MC) simulated events to train event classifiers, create analysis categories, and build signal models. These simulated events go through the same processes as data; this includes being "measured" through a simulated CMS detector, the reconstruction process using the PF algorithm, and the rest of the analysis chain.

Several Higgs boson production modes have been simulated at next-to-leading order (NLO) in perturbative QCD for H$\rightarrow \gamma\gamma$ analyses using the MAD-GRAPH5_AMCNLO [**?**] in combination with PYTHIA8 [**?**] (to perform parton showering and hadronization) and

POWHEG generators [**?**]. Events from the ggH production mode are further weighted to agree with next-to-next-to-leading order programs (NNLOPS) [**?**]. The events produced using the POWHEG generator are exclusively used to train event classifiers for the analysis to ensure that events used in training are not used in this analysis.

Despite being excluded from the background model, simulated background events are used to train event classifiers. The dominant background to H$\rightarrow \gamma\gamma$ analyses is the irreducible SM diphoton production, predominantly from $gg \rightarrow \gamma\gamma$. This background is simulated using the SHERPA program [**?**]. The other types of background are γ+ jet (prompt-fake) events and jet + jet (fake-fake) events where the jet "fakes" a photon signature in the detector. These backgrounds are modeled using PYTHIA8, and selection criteria are applied to ensure a high density of events with large ECAL energy fractions since jets with large ECAL energy fractions look more like fake photons than those with larger HCAL energy fractions.

All simulated events pass through a simulated version of the CMS detector built in GEANT4 [**?**]. Additional pile-up interactions are included in the simulation. When necessary, the distribution of the number of pile-up interactions in the simulation is reweighted to match that of the corresponding year of data.

4.4 Photon reconstruction

The H$\rightarrow \gamma\gamma$ analysis uses photon candidates selected by the PF algorithm. The energy of these candidates is determined by the associated ECAL SC. It is common for photons to interact upstream of the detector, becoming a *converted* photon or rather an electron-positron pair. A multivariate regression (MVA) is used to correct for energy losses upstream of the SC to estimate the initial energy of the photon. This MVA is trained on simulated data and instructed to use simulated detector readouts to predict the true energy of a simulated photon. After the MVA correction, residual energy scale corrections are applied to the data, and additional smearing corrections are applied to the simulation. Details on this procedure are presented in Chapter **??**. Once the photon energy has been corrected,

selection criteria are applied to pick diphoton events most likely to have come from a Higgs boson. These selection criteria contain kinematic cuts and cuts on the photon identification boosted decision tree (BDT) score to distinguish prompt from fake photons. The full suite of selection criteria are described below.

The variables used in the reconstruction process and their definitions are all listed in Table **??**.

4.4.1 Photon energy

The energy response of the ECAL (E_{true}/E_{SC}) is characterized by a Gaussian core with power law tails. The algorithm used to estimate the energy of an SC is shown in Equation **??**. In this equation, the term $F_{e/\gamma}$ corrects the SC energy to the energy of the e/γ and accounts for energy losses from incomplete shower containment due to early conversion, energy lost to the gaps between ECAL crystals and modules, energy lost at the back of the crystal from a long electromagnetic shower or late initial interaction in the crystal, and energy lost to noisy crystals in the clustering algorithm. This correction is derived using an MVA regression that learns the shape of the ECAL energy response and provides a predicted PDF of the ECAL energy response for each photon.

After the MVA regression corrections, additional corrections are applied to the photon energy to account for residual differences between data and MC. The motivation and derivation of these corrections are described in Chapter **??**.

4.4.2 Photon pre-selection

Addition selection criteria are applied to diphoton events to reduce the number of events in which one or two jets fake a photon. These criteria are as follows:

1. Minimum p_T: the leading photon (most energetic) is required to have a $p_T > 35$ GeV, and the subleading photon (least energetic) is required to have a $p_T > 25$ GeV.

2. Geometric acceptance: both photons must fall within $|\eta| < 2.5$, excluding the transition region $1.4442 < |\eta| < 1.566$.

3. Electron Veto: both photons must have a True electron veto, i.e., neither photon can have an associated track.

4. Hadronic shower rejection: The fraction of energy deposited in the HCAL to that of the ECAL SC cannot exceed 0.08 ($H/E_{SC} < 0.08$).

5. Shower shape and isolation requirements:

   (a) Photons in the barrel: Photons with $R_9 > 0.85$ have no additional isolation requirements. Photons with $0.5 < R_9 < 0.85$ must have $\sigma_{i\eta i\eta} < 0.015$, a photon isolation ($\mathcal{I}_{\text{ph}}$) < 4.0 GeV, and a tracker isolation ($\mathcal{I}_{\text{tk}}$) < 6.0 GeV.

   (b) Photons in the endcap: Photons with $R_9 > 0.90$ have no additional isolation requirements. Photons with $0.8 < R_9 < 0.9$ must have $\sigma_{i\eta i\eta} < 0.035$, a photon isolation ($\mathcal{I}_{\text{ph}}$) < 4.0 GeV, and a tracker isolation ($\mathcal{I}_{\text{tk}}$) < 6.0 GeV.

   (c) All photons must have $R_9 > 0.8$, a charge isolation ($\mathcal{I}_{\text{ch}}$) < 20 GeV, and a charge isolation to transverse momentum ($\mathcal{I}_{\text{ch}}/p_T^{\gamma}$) < 0.3.

The pre-selection efficiency of these criteria, excluding the electron veto, is calculated using the tag-and-probe method applied to Z$\rightarrow ee$ events. The electron veto efficiency is calculated using Z$\rightarrow \mu\mu\gamma$ events. These events must have an invariant mass of 60 to 120 GeV. The pre-selection criteria, except the electron veto, are then applied to the photons, and the efficiency of the electron veto is found from the fraction of events that pass the electron veto. The pre-selection efficiency is around 90% and rises above 95% for photons with $R_9 > 0.95$. Per photon weights match the pre-selection efficiency in simulation to that observed in data.

4.4.3 Photon identification

A photon identification BDT is used to reject fake photons and is trained on a γ+jets simulation sample. This BDT takes as input the shower-shape, kinematic, and isolation variables, as well as variables sensitive to pile-up, such as ρ. One of the historically larger sources of uncertainty in H$\rightarrow \gamma\gamma$ analyses is the modeling of EM showers in simulation. Since the

BDT takes several shower-shape variables as input, any discrepancies in the shower-shape variables between data and simulation will be propagated to the output of the BDT, significantly affecting its performance. These discrepancies are addressed by correcting the shower shape variables used as input to the photon ID BDT using a chained quantile regression (CQR) [**?**]. Chapter **??** provides additional details on the systematic uncertainty.

For a given input variable, the CQR method trains a series of 21 BDTs to predict the input variable's cumulative distribution function (CDF) at 21 points along its range. For example, a series of BDTs predict the CDF(R_9) at the points [0.01, 0.05, 0.10, ..., 0.95, 0.99]. Once the predicted values are obtained, they are linearly interpolated to get the full CDF. This process is performed on both data and simulation to define a correction value for each electron in simulation such that the input variable matches the shape of the variable in data. The correction is defined as:

$$S_i \rightarrow S_i^{\text{corr}} = \text{CDF}_{\text{data}}^{-1}(\text{CDF}_{\text{simulation}}(S_i)) \tag{4.2}$$

where S_i is any variable used as input to the photon ID BDT.

This method's "chained" aspect refers to the inclusion of previously corrected shower shape variables in the training of successive CDF BDTs. Consider a set of N shower shape variables [S_1, S_2, S_3, ..., S_N]. The BDTs trained to predict points along CDF(S_1) are given only p_T^e, η, ϕ, and ρ. However, the BDTs trained to predict points along CDF($S_i|_{i\neq 1}$) are given the initial variables, as well as the corrected shower shape variables [S_1, ..., S_{i-1}] (or [S_1^{corr}, ..., S_{i-1}^{corr}] in the case of simulation). This chaining of the input variables accounts for their correlations, and the order of the chaining is optimized to reduce differences in the photon ID BDT output in data and simulation.

Isolation variables are treated slightly differently due to the discontinuity between the peaks and tails of the distributions. Two classifiers are trained to predict whether an electron should fall in the peak or tail of the given distribution, and simulated electrons are moved between the peak and tail based on the output of the classifiers so that the composition of the tail and peak match that of data. After this process, the CQR method is applied to

electrons in the tails of the isolation variables. Ref [**?**] provides additional information on the CQR method and its applications.

Finally, a loose cut is applied to the photon ID BDT score (> -0.9) to reduce the number of γ+jet and jet+jet events by 70% while keeping $> 99\%$ of signal events. This cut ensures that the statistical power of the background simulation is strong enough to train event categorization discriminants. The distribution of the photon-ID BDT score for the lowest scoring photon in diphoton events, as well as the same for electrons reconstructed as photons, is shown in Figure **??**

4.5 Vertex reconstruction

Equation **??** shows the relationship between the diphoton invariant mass and its relationship to the photon energies and their opening angle. Recall the claim that keeping the resolution on the diphoton vertex to less than 1 cm ensures that the dominant sources of uncertainty on the invariant mass come from the photon energies. This section explains the origins of this claim.

To begin, recall that there are multiple p-p collisions in each bunch crossing, meaning multiple vertices are spread along the beam line's axis. Suppose the wrong vertex is chosen from among roughly the 35 vertices. In that case, the opening angle of the two photons directly affects the diphoton invariant mass resolution. This effect is negligible when the chosen vertex is within 1 cm of the true vertex, and to ensure this resolution on vertex selection is achieved, a BDT is trained to identify the most probably primary vertex.

4.5.1 Vertex selection

The BDT trained to select the most probable primary vertex is trained on simulated ggH H$\rightarrow$ $\gamma\gamma$ events and takes the following variables as input. The sums in the following definitions run over the PF tracks associated with a given vertex with label i, and the quantity $\overline{p}_T^{\gamma\gamma}$ is the transverse momentum of the diphoton system with respect to the vertex i.

- $\sum_i |\overline{p}_T^{(i)}|^2$

- $-\sum_i \frac{\vec{p}_T^{(i)} \cdot \vec{p}_T^{\gamma\gamma}}{|\vec{p}_T^{\gamma\gamma}|}$
- $\frac{|\sum_i \vec{p}_T^{(i)}| - p_T^{\gamma\gamma}}{|\sum_i \vec{p}_T^{(i)}| + p_T^{\gamma\gamma}}$

For events with photon conversions, the BDT is given the following inputs in addition to those above:

- the number of photon conversions in the tracker
- $\frac{|z_{\rm vtx} - z_{\rm e}|}{\sigma_z}$: the pull between the z-positions of the reconstructed vertex ($z_{\rm vtx}$) and the estimated vertex from photon conversions ($z_{\rm e}$) with respect to the uncertainty on $z_{\rm e}$ (σ_z).

The BDT creates a score for each vertex; the vertex with the highest score is chosen as the primary interaction point. If the primary interaction point is within 1 cm of the actual interaction point, it is labeled as the "correct" vertex. The vertex choice efficiency in H$\rightarrow \gamma\gamma$ events is $\sim$79%, shown in Figure **??** alongside the vertex probability BDT score. The vertex probability BDT is trained to predict the probability that the selected vertex is within 1 cm of the true diphoton vertex. This BDT takes as input the number of reconstructed vertices, the relative positions and vertex assignment BDT scores for the three highest-scored vertices, the transverse momentum of the diphoton system, and the number of converted photons in the tracker. This BDT is trained on simulated ggH, H$\rightarrow \gamma\gamma$ events.

4.6 Electron reconstruction

While the differences between photon and electron reconstruction are minor, they are essential to note. During reconstruction, electrons must have a matching track instead of a track veto. Additionally, electrons can be seeded by either tracks or calorimeter deposits. The electron energy regression includes information about tracks and accounts for energy lost to bremsstrahlung. Ref [**?**] contains more details on electron reconstruction.

Chapter 5

Measurement of the energy scale and resolution of the CMS ECAL

5.1 Introduction

The precise calibration of the CMS ECAL is essential in making a precision mass of the Higgs boson measurement in the diphoton channel. This chapter focuses on the calibration of the ECAL, with detailed descriptions of the setting of the energy scale and the procedure for matching the resolution in simulation with the observed resolution in the data.

First, the need for residual energy scales and additional Gaussian smearings is motivated by discussing various detector features that affect the energy scale and resolution. Then, these detector properties are summarized by measurements of Z$\rightarrow$ *ee* events. A method of correcting the residual differences in energy scale and resolution between data and simulation was developed in 2012 and is described in Ref [**?**]. From this method, a modern technique to correct these residual differences was developed that relies on the increased data in Run 2. Residual energy scale corrections and additional smearings from this new technique are incorporated in the H$\rightarrow$ $\gamma\gamma$ analysis. Finally, the results of the method are demonstrated through the agreement of the dielectron invariant mass distribution in data and simulation.

5.2 The energy resolution of the CMS ECAL

The intrinsic energy resolution of the CMS ECAL, as measured in a test beam, takes the form:

$$\frac{\sigma(E)}{E} = \frac{2.8\%}{\sqrt{E}} \oplus \frac{12\%}{E} \oplus 0.3\% \tag{5.1}$$

where E is the energy of an interacting particle in GeV, and σ is the energy resolution. The three components represent, in order, the stochastic term due to fluctuations in the electromagnetic shower development, the noise term due to electronic noise, and the constant term due to a lack of full containment of the shower. These terms form the basis of the energy resolution of the CMS ECAL. However, additional factors affect the energy scale and resolution of the CMS ECAL, including the stability of the time response corrections, the precision of the inter-calibration corrections, and interactions with upstream material.

5.2.1 Stability of the time response corrections

The scintillation response of an ECAL crystal is related to its laser response via:

$$\frac{S(t)}{S_0} = \left(\frac{R(t)}{R_0}\right)^{\alpha} \tag{5.2}$$

where $S(t)$ is the scintillation response of the crystal at time t, S_0 is the initial scintillation response of the crystal, $R(t)$ is the laser response of the crystal at time t, R_0 is the initial laser response of the crystal, and α is a material dependent quantity. The value of α encodes both properties of the crystals themselves, such as roughness, transition, and manufacturing quality, as well as properties of the photodetectors used to read the scintillation light from the crystals, such as dark currents in the APDs, and the response loss of the VPTs due to irradiation. The value of α is fixed in the EB based on average values measured by monitoring the drift of the observed mass of the π^0. In the EE, α is derived as a function of location in the detector to account for the effect of increased crystal irradiation at high values of $|\eta|$ by monitoring the time stability of the mass of the Z boson. These corrections work well enough to ensure the relative energy resolution on the mass of the Z boson is around

0.1% in the EB and between 0.1% and 0.3% in the EE. Since they are time-dependent, these corrections are applied only to data. Simulation does not account for any effects due to the irradiation of ECAL crystals. Therefore, the relative energy response in simulation is smaller than in data.

5.2.2 Inter-calibration precision

Due to natural variations in the manufacturing process, each crystal's response in the ECAL is slightly different. These variations in the response are corrected using a set of corrections known as inter-calibrations.

To derive these inter-calibrations constraints from $\pi^0/\eta \to \gamma\gamma$ events, momentum constraints from Z and W decays, and the Z mass constraint in $Z \to ee$ decays are used to align the response and resolution of each crystal in the ECAL. The precision of the inter-calibrations for the Run 2 datasets is less than $\sim$0.5% in the EB and less than 1% in the EE. Variation with η for the different years of data taking is shown in Figure **??**. The inter-calibration precision contributes to the constant term of the energy resolution and is included in the simulation as a Gaussian mis-calibration to the crystal response.

5.2.3 Interaction with the upstream material

It is essential to understand the amount of material between the ECAL and the interaction point for precise energy and resolution calibration. Electromagnetic objects may interact with this upstream material, beginning their shower development before reaching the ECAL. The amount of material between the ECAL and the interaction point, as implemented in the simulation of the CMS detector, is shown in Figure **??**. In the case of an upstream interaction, some energy will inevitably be lost before the ECAL. The shower shape variable R_9 is the ratio of energy deposited in the core of an electromagnetic shower to the total energy measured from the shower. It reflects how strongly an electron or photon has interacted upstream of the ECAL and, thus, how much energy it has lost in transit. A threshold at $R_9 = 0.96$ is set to divide low-bremsstrahlung objects from high-bremsstrahlung objects, or those

without significant upstream interactions and those with substantial upstream interactions. Though good, the simulation of the ECAL is not perfect, and differences in the model of the upstream material and the ECAL itself contribute to residual differences in the energy resolution of data and simulation.

5.3 The energy scale and resolution of electrons in the CMS ECAL

The energy scale and resolution of objects in the CMS ECAL have been described now as a function of time, position in the detector of the reconstructed object, and the shower shape variable R_9 associated with the object. These effects are monitored closely, and the stability of the scale of reconstructed Z bosons and the stability of the R_9 of the leading electron in Z$\rightarrow ee$ events are shown in Figures **??** and **??** for events across all of Run 2.

The relative energy resolution of the CMS ECAL in Run 2 measured through Z$\rightarrow ee$ decays is shown in Figure **??** for the case low bremsstrahlung electrons and the inclusive case.

There are residual differences between data and simulation in both the scale and the width of the dielectron invariant mass distributions, even after reconstruction and initial corrections. Examples of the agreement after reconstruction are shown in Figure **??**.

5.3.1 Z$\rightarrow e^+e^-$ event selection

Trigger

The trigger menu cuts are described in Table **??** and are identical to the diphoton events cuts.

Offline selection

Events used in this analysis are required to have passed the selected trigger, and dielectron events must fulfill the following additional requirements:

- Leading and subleading electrons must have an $E_T > 32$ and $E_T > 20$ respectively
- Both electrons must pass the loose electron identification and isolation criteria
- Both electrons must be in the ECAL fiducial region, with both electrons in the tracker acceptance ($|\eta| < 2.5$), with a veto applied to electrons near the barrel transition region, meaning $1.4442 < |\eta| < 1.566$.

Events must also satisfy cuts on kinematic variables in the CMS standard electron identification criteria:

- $\Delta\eta$: The difference in pseudorapidity between the measured track associated with the electron and the vertex and the center of the ECAL cluster extrapolated to the vertex.
- $\Delta\phi$: The difference in the angle in transverse with respect to the beam axis between the measured track of an electron at the vertex and the center of the ECAL cluster extrapolated to the vertex.
- $\sigma_{i\eta i\eta}$: A shower shape variable defined as

  $$\sigma_{i\eta i\eta} = \sqrt{\frac{\sum(\eta_i - \overline{\eta})^2 w_i}{\sum w_i}}, \text{where } \overline{\eta} = \frac{\sum w_i \eta_i}{\sum w_i} \text{ and } w_i = \max\left(0, 4.7 + \log\left(\frac{E_i}{E_{5\times5}}\right)\right)$$

  where the sum runs over the 5×5 crystal matrix around the crystal in the cluster containing the largest deposited energy, and the η distances are measured in units of the size of the crystals in the η direction.
- H/E: The ratio of the energy deposited in the HCAL in a cone of $\Delta R = 0.15$ centered around the seed of the ECAL cluster to the energy deposited in the ECAL.
- Impact Parameters: The dielectron vertex position is compared with the closer primary vertex in the transverse plan with respect to the beam axis ($d0(vtx)$) and in the beam axis direction ($dz(vtx)$).
- $|1/E - 1/p_{in}|$: The absolute difference between the inverse of the supercluster energy and the inverse of the track momentum at the vertex.

- PFiso: Based on the particle flow algorithm, the isolation variable is defined as a combination of the activity of charged and neutral particles around the considered electron after removing the electron deposit.

- Conversion rejections: Two cuts are used to reject converted photons: the vertex fit probability must be below 1e−6, and the number of missing points in a reconstructed track is constrained to be one or fewer.

The criteria for each of these variables are defined for different efficiencies and purities and are summarized in Table **??**.

The primary sources of background found in these events come from electroweak processes like $W \to \ell\nu$, and $Z \to \tau\tau$, and QCD events, including $t\bar{t}$ events. The loose working point has an efficiency of around 90% and contains negligible background.

5.3.2 Simulation

Two large simulated MC samples are used for the $Z \to ee$ signal. These are summarized in Table **??**.

5.3.3 Comparison of data and MC samples

Previous studies have shown differences in the shapes of the $p_T(Z)$ and rapidity ($y(Z)$) distributions in data and the above simulation. These differences are corrected using weight factors derived in bins of $p_T(Z)$ and $y(Z)$. The weights ensure that the shapes of these distributions match. First, a 2D histogram of event counts is created in the $p_T(Z)$ and $y(Z)$ space in both simulation and data. The histograms are normalized to preserve the total event count after reweighting. The weights are calculated by dividing the normalized number of events in the data by the normalized number of events in the simulation. Then, by applying these weights to the simulation, the shape of $p_T(Z)$ and $y(Z)$ will match that in the data. The bins in which the scale factors were derived are:

- If $p_T(Z) < 40$ GeV then the bin sizes are 1 GeV, otherwise $[40, 45, 50, 55, 60, 80, 100, > 100]$
- $y(Z)$: $[0, 0.3, 0.6, 0.9, 1.2, 1.5, 1.7, 1.8, 2.1, 2.3, 2.5]$, and similarly for negative $y(Z)$.

5.4 The global smearing method

The global smearing method (GSM) aligns the scale of the dielectron invariant mass peak in data with the scale of the dielectron invariant mass peak in simulation and applies additional Gaussian smearing to the dielectron invariant mass distribution in simulation to match the resolution of the same distribution in data. This method is described in detail in this section.

5.4.1 Framework outline

The GSM framework is a suite of tools built in Python whose primary purpose is to improve the agreement in data and simulation by minimizing a loss function built around the negative log-likelihood, and reducing the difference between data and simulation in categories defined in terms of single electron properties. SciPy's Optimize package handles the minimization and implements the L-BFGS-B method: a limited memory, bounded, hill climbing algorithm that is robust and fast for large phase spaces [**?**, **?**, **?**]. The general outline of the framework, with more details on each step provided in subsequent sections below, is as follows:

- define n scale categories and k smearing categories based on single electron properties such as location in the detector ($|\eta|$), the probability of having had an upstream interaction (R_9), and transverse energy (E_T).
- Associate a scale with each scale category n, and a Gaussian width with each smearing category k.
- Construct $n(n+1)/2$ dielectron invariant mass distributions in both data and simulation where each distribution contains events with electrons from two, and only two, single electron scale categories n and m, and two smearing categories k and l.

- Evaluate a loss function across all dielectron invariant mass categories ((*n*,*j*),(*k*,*l*))
- Provide the loss function to SciPy's minimize tool in the Optimize sub-library and allow the minimizer to find values of all n scales and all k Gaussian widths for the smearing categories.

The framework improves the agreement between data and simulation in a series of compounded steps, with the binning becoming progressively finer. The different steps are as follows:

- Run number $\times$ $|\eta|$: The first step uses the "median method" (described in detail below) to correct the median invariant mass to that in simulation in bins of run number, and $|\eta|$ which stabilizes the dielectron invariant mass distribution in time and location in the detector.
- $|\eta| \times R_9$: The next step uses the GSM's minimization procedure to derive both the residual scales and additional smearings simultaneously in bins of $|\eta|$ and R_9 using electrons that have already been corrected using the scales derived in the previous step. This compounding effect further improves the agreement between data and simulation. From now on, this compounded process will be referred to as "on top of", i.e., a new set of scales is derived *on top of* a previous set of scales. The bins are coarse, with four bins in $|\eta|$, as in the last step, and two bins in R_9, where the divide between bins separates low-bremsstrahlung and high-bremsstrahlung electrons.
- Granular $|\eta| \times R_9$: The minimization procedure is used to derive the residual scales and additional smearings simultaneously in bins of $|\eta|$ and R_9. In this step, the smearings are rederived as the bins remain coarse with four bins in $|\eta|$ and two bins in R_9. The granular $|\eta| \times R_9$ scales are derived on top of the $|\eta| \times R_9$. The bins sizes are much more granular than in the previous step, with five bins in $|\eta|$ and between 8 and 16 bins in R_9.
- $|\eta| \times E_T$: In this step, a set of residual scales are derived on top of the results from the previous step in 5 bins of $|\eta|$ and between 4 and 10 bins in electron E_T.

- $|\eta| \times E_T$: This step derives a set of residual scales *and* additional smearings in 4 bins of $|\eta|$, two bins of R_9, and four bins of electron E_T. The scales are derived on top of the previous results.

5.4.2 The "median method"

The "median method" is used to derive the first set of corrections, which stabilize the dielectron invariant mass distribution in time and location in the detector. Dielectron invariant mass distributions are built for each run bin, defined as a group of runs with a minimum of 10,000 dielectron events, and are divided into four bins in $|\eta|$. The scale for each category is the factor that corrects the median of the dielectron invariant mass distributions to the PDG value of the mass of the Z boson:

$$\text{scale}^{\text{run bin}} = 91.188 \text{ GeV}/\text{median}(M_{ee}^{\text{run bin}}). \tag{5.3}$$

5.4.3 Dielectron event categorization

The GSM is used to derive corrections to single electron energies by scaling the energy of electrons in data to match the known value of the Z mass of 91.188 GeV [**?**] in simulation, and to derive an additional Gaussian smearing to be applied to electrons in simulation to match the resolution in data. The GSM uses simulated Z boson decays as a template PDF for the data. This template approach accounts for known detector effects, reconstruction inefficiencies, and the kinematic behavior of the Z boson.

The categorization of the dielectron invariant mass is defined by n categories based on single electron properties. This categorization approach produces $n(n+1)/2$ dielectron invariant mass categories (D for data categories and S for simulation categories). Each category is associated with two electron energy scales (ΔP_n, ΔP_m) applied to the energy of electrons in data and two electron energy smearings ($\Delta\sigma_k$, $\Delta\sigma_l$) applied to the energy of electrons in simulation ($D_{n,m}$ and $S_{n,m}^{k,l}$).[1] The smearings are a multiplicative factor obtained

[1] It is essential to note that all scale categories (n, m) are contained in the broader smearings categories (k, l) and that both data and simulation are categorized by (n, m)

from a random sampling of a Gaussian distribution of width $\Delta\sigma$ centered around 1, and cause the width of the dielectron invariant mass distribution in simulation to increase, since the width of the dielectron invariant mass distribution in simulation is always less than the width of that in data, by construction. The invariant mass distributions in the simulation are over-sampled to smooth the distributions after applying the additional Gaussian smearings and to increase the sensitivity of the additional smearings impact on the loss function. Over-sampling is implemented by doubling the number of events in simulation in a given category until the number of events exceeds 50 times the number of events in data or 50,000 events, whichever is larger. If there is a very high statistics category in data, an upper limit on the over-sampling of simulation is set at 10^6 events to reduce the computational time of the method.

Once all categories are constructed, low statistics categories are deactivated. In data, active dielectron categories must have more than ten events. In simulation, active diagonal ($n = m$) dielectron categories must contain at least 1000 events, and all active off-diagonal ($n \neq m$) dielectron categories must contain at least 2000 events.

5.4.4 Dielectron invariant mass binning

The loss function used to evaluate the agreement between data and simulation is an event-weighted, binned, negative log likelihood that is combined with the chi-squared test. The dielectron invariant mass histograms are built for data and simulation in each dielectron category constructed from single electron categories n and m. The number of bins in each dielectron category is determined using the Freedman-Diaconis rule, which is defined as:

$$N_{n,m} = min(\left\lfloor \frac{D_{n,m}^{1/3}}{2 * IQR(M_{ee}^{data,n,m})} \right\rfloor, \left\lfloor \frac{S_{n,m}^{1/3}}{2 * IQR(M_{ee}^{sim,n,m})} \right\rfloor) \tag{5.4}$$

where $D_{n,m}$ is the number of events in the data dielectron invariant mass histogram, IQR is the inter-quartile range, $M_{ee}^{data,n,m}$ is the distribution of dielectron invariant mass events in category (n, m) in data, $S_{n,m}$ is the number of events from the dielectron invariant mass histogram in simulation, and $M_{ee}^{sim,n,m}$ is the distribution of dielectron invariant mass events

in category (n, m) in simulation. By default, each histogram is truncated, with a lower bound at 80 GeV and an upper bound at 100 GeV. If the tails of the dielectron invariant mass histograms in data and simulation show a large disagreement that skews the results, the bounds on the histograms are made tighter to allow the fit to focus on improving the agreement between data and simulation in the core of the dielectron invariant mass distribution.

5.4.5 Loss function

The binned negative log-likelihood between data and simulation is defined as follows:

$$NLL_{n,m} = -2 * \sum_{i=1}^{N_{n,m}} d_i p_{n,m} log(m_i \sigma_{n,m}) + (D_{n,m} - d_i p_{n,m}) log(1 - m_i \sigma_{n,m}) \tag{5.5}$$

where

$$\sigma_{n,m} = \sqrt{(}Gaus(1, \Delta\sigma_n) Gaus(1, \Delta\sigma_m)), \text{ and } p_{n,m} = (1 + \Delta P_n)(1 + \Delta P_m). \tag{5.6}$$

In these equations, n and m represent the single electron category indices, $N_{n,m}$ is the number of bins in the histograms of data and simulation in category $(n,\ m)$, d_i is the content of bin i in the data histogram for the category (n, m), m_i is the content of bin i in the normalized histogram in simulation for the category (n, m), $p_{n,m}$ is the factor containing the residual scales associated with the single electron categories n and m, $\sigma_{n,m}$ is the factor containing the additional smearings associated with single electron categories n and m, $D_{n,m}$ is the number of events in the histogram of the dielectron invariant mass in category (n,m). After deactivating categories with too few events, a one-dimensional scan of the loss function for all dielectron categories associated with the single electron category is used to determine an initial scale value in each single electron category. The value that minimizes this scan over the loss function is chosen as the initial value of the scale for each single electron category. After initializing the values of the residual scales, initial values for each additional smearing are selected similarly.

The global NLL is the product of the weighted average of the NLL and the reduced chi-squared between data and MC for all active dielectron invariant mass categories:

$$NLL_{global} = \frac{\sum_{n=1}^{N}\sum_{m=i}^{N} w_{n,m} * NLL_{n,m} * \chi^2_{n,m}}{\sum_{n=1}^{N}\sum_{m=1}^{N} w_{n,m}} \tag{5.7}$$

where the weight $w_{n,m}$ is 1, if the category is diagonal ($n = m$), and 0.01 if the category is off-diagonal ($n \neq m$). Prioritizing diagonal categories reduces the influence of categories with few events and enhances the quality of the fit in categories crucial to analyses using the GSM. The NLL_{global} is given as input to the minimization function in SciPy's Optimize package. Rectangular bounds on each scale and smearing are enforced at ± 0.05. These boundaries prevent the minimizer from finding non-physical phase spaces, which may inadvertently minimize NLL_{global}. In minimization steps where the single electron categories are E_T dependent, the boundaries are further restricted to ± 0.01 to prevent non-physical solutions and improve computation time.

5.4.6 Validation of the GSM

The method was validated by testing its ability to retrieve injected values of scales and smearings. Two studies were performed: a scale only test, and a scales and smearings test. For these studies, the simulated Z boson decays used as data were not over-sampled, however the over-sampling requirement was applied to samples used as simulation.

In the first study, an artificial scale was injected in each single electron category. Smearings were not injected in this study. The injected scale was assigned at random in the range $\pm 1\%$. The GSM was found to determine the scale to within 0.01% of the injected value in all categories using coarse bins of $|\eta|$ by R_9.

In the second study, both scales and smearings were artificially injected. The scales and smearings were assigned at random in the range $\pm 1\%$. It was found that the injected scales were recovered to within 0.25% of the injected value, and most scales were recovered to within 0.1% of the injected value. In contrast, the smearings were recovered at between 0.25% and 3.4% of the injected values. This larger value was due to the relatively flat shape of the NLL around the true minimum in the smearings phase space.

5.5 Results

The results of the GSM and its ability to improve the agreement of data and simulation as measured through the dielectron invariant mass peak are summarized below.

5.5.1 Time stabilization

Before any corrections, the median of the dielectron invariant mass distribution in data wanders significantly across a year of data due to the effects mentioned above. The median invariant mass as a function of the run number is shown in Figure **??**.

As described above, we apply the “median method" to the data to correct the median dielectron invariant mass to the PDG value of Z mass. The corrected median dielectron invariant mass is shown for the 2016, 2017, and 2018 datasets in Figure **??**.

5.5.2 E_T dependent residual scales and additional smearings

After applying the run-dependent corrections, the residual scales and additional smearings are derived simultaneously using the “global smearing method" in 10-13 bins in R_9 and 5 in $|\eta|$. By binning finely in R_9, the corrections become independent of the effective R_9 distributions in this analysis' categories. These corrections for the 2017 dataset are shown for the EB and EE in Figure **??**. The smearings for this set of corrections are derived in coarser bins of $|\eta|$ and R_9 with four bins in $|\eta|$ and two bins in R_9.

In Run 1, a set of E_T dependent smearings was derived to correct for differences in resolution as a function of the energy of the electrons. These smearings were derived by decoupling the constant term smearing from the stochastic term smearing using an angle ϕ. Using the GSM, we derive the stochastic term as a constant term with constraints on E_T. This approach yields stable stochastic smearings, which are more stable than the Run 1 approach. The increased stability also comes with smaller uncertainties on the stochastic smearings. A set of scales and stochastic smearings were derived for the electron energy using the GMS in four bins of $|eta|$, four bins of E_T, and two bins of R_9. These scales and smearings were derived to account for the differences in resolution as a function of energy

and have yielded greatly improved agreement in data and simulation. The final smearings for 2016, 2017, and 2018 are listed in Table **??**.

The systematic uncertainty associated with the scales and smearings that account for the variations in R_9, electron identification, pre-selection E_T threshold, and non-closure are all added in quadrature. The details of this study are given in Chapter **??**. The agreement between data and MC after all corrections is shown for electrons with $R_9 > 0.96$ in the barrel-barrel and barrel-endcap categories for the 2016 preVFP dataset in Figure **??**, the 2016 postVFP dataset in Figure **??**, the 2017 dataset in Figure **??**, and the 2018 dataset in Figure **??**.

Chapter 6

Event categorisation

6.1 Overview

After the H$\rightarrow \gamma\gamma$ events are reconstructed, and before categorization, they are subject to additional cuts on their transverse momenta to improve the agreement between data and simulation in the lower tails of the diphoton invariant mass distribution. The event must have a diphoton invariant mass in the range 100 GeV to 180 GeV, and the leading (highest energy) and subleading (second highest energy) photons must satisfy $p_T^{\gamma 1} > m_{\gamma\gamma}/3$ and $p_T^{\gamma 2} > m_{\gamma\gamma}/4$ respectively. Additionally, all events must be within $|\eta| < 2.5$, but not in the ECAL barrel-endcap transition region $1.4442 < |\eta| < 1.566$.

All diphoton events are sorted into categories based on their mass resolution and predicted signal-to-background ratio using the diphoton BDT score (described below). This sorting, or categorization, enhances the sensitivity of the analysis and improves the overall precision of the Higgs boson mass measurement. Unlike previous studies, this analysis does not categorize based on production mode. Adding additional production mode categories results in a slight improvement in the precision of the mass measurement by reducing the statistical uncertainty. However, since the precision of this analysis is limited by the systematic uncertainty, small gains in the statistical uncertainty are negligible.

6.2 Diphoton BDT

The diphoton BDT is an event classifier that assigns high scores to events with the following properties:

- the photons have signal-like momentum and energy distributions
- the photons have sufficiently high photon ID BDT scores
- a good diphoton mass resolution

The diphoton BDT does not use the invariant mass directly, to avoid biasing the results. The diphoton BDT takes the following variables as input:

- the transverse momentum rescaled by the invariant mass of the leading (1) and subleading (2) photons: $p_T^{\gamma 1(2)}/m_{\gamma\gamma}$
- the pseudorapidity of both photons: $\eta^{1(2)}$
- the cosine of the angle between the photons: $\cos(\theta_{1,2})$
- the photon ID BDT score for both photons
- the relative mass resolution, assuming reconstruction happened under the right vertex (RV): σ_{RV}
- the relative mass resolution, assuming reconstruction under the wrong vertex (WV): σ_{WV}
- the vertex probability BDT score: p_{vtx} (see Sec **??**).

In the above definitions, the RV relative mass resolution assumes Gaussian resolution functions for the photon energy and is defined as:

$$\sigma_{\mathrm{RV}} = \frac{\sigma_m^{\mathrm{right}}}{m_{\gamma\gamma}} = \frac{1}{2}\sqrt{\left(\frac{\sigma_{E_1}}{E_1}\right)^2 + \left(\frac{\sigma_{E_2}}{E_2}\right)^2}. \tag{6.1}$$

Under the WV assumption, the relative mass resolution includes an extra factor that measures the distance between the correct vertex and the selected primary vertex: σ_{vtx}. This extra factor leads to the following definition:

$$\sigma_{\text{WV}} = \frac{\sigma_m^{\text{wrong}}}{m_{\gamma\gamma}} = \sqrt{\left(\frac{\sigma_m^{\text{right}}}{m_{\gamma\gamma}}\right)^2 + \left(\frac{\sigma_{\text{vtx}}}{m_{\gamma\gamma}}\right)^2}. \tag{6.2}$$

The following weights are applied to signal events used to train the BDT:

$$w_{\text{sig}} = \frac{p_{\text{vtx}}}{\sigma_{\text{RV}}} + \frac{1 - p_{\text{vtx}}}{\sigma_{\text{WV}}}. \tag{6.3}$$

These weights ensure that events with a better mass resolution are assigned a higher diphoton BDT score because they are suppressed by σ_{RV} and σ_{WV}, which get smaller as the relative resolution improves. The diphoton BDT score for the full Run 2 dataset is shown in Figure **??**.

6.3 Event categorization using the diphoton BDT score

Splitting the analysis into categories where events have similar properties, increases the expected significance and improves the precision of the analysis. The categories are defined in terms of the boundaries in the diphoton BDT space. Their location is determined by minimizing a loss function that targets the relative diphoton invariant mass resolution ,and the ratio of signal events to the square root of background events ($\text{S}/\sqrt{\text{B}}$). The number of categories used in the analysis is determined by eye by comparing the degree of improvement between the minimum of the loss function for n categories and $n+1$ categories.

6.3.1 Samples

The signal samples used in this derivation are NLO samples with the ggH, VBF, VH, and ttH production modes generated with a Higgs mass set at 125 GeV. The background samples are generated with a data-driven approach outline in Section 6.5 of [**?**].

6.3.2 Boundary optimization and loss function

The optimization process begins by picking a random set of n boundaries in the range 0 - 1, whose values are denoted as b_i. To ensure that n boundaries correspond to n categories and not $n+1$, all events in the range 0 - b_0 are ignored, leaving only events in the range b_0 - 1. For each category c_i, with lower bound b_i, the minimum range in the diphoton invariant mass space containing 68.3% of events is computed. Events in this range are used to calculate the standard deviation ($\sigma_i^{\gamma\gamma}$) and mean ($m_i^{\gamma\gamma}$) of the invariant mass as well as their uncertainties.

The loss function used in the optimization of the diphoton BDT score boundaries is defined as:

$$L = 1000 * \frac{\sum_i \sigma_i^{\gamma\gamma}/\sigma_{\sigma_i^{\gamma\gamma}}}{\sum_i 1/\sigma_{\sigma_i^{\gamma\gamma}}} \frac{\sum_i 1/\sigma^2_{m_i^{\gamma\gamma}}}{\sum_i m_i^{\gamma\gamma}/\sigma^2_{m_i^{\gamma\gamma}}} * \sum_i \frac{\mathrm{B}_i}{\mathrm{S}_i^2}. \tag{6.4}$$

Where $\sigma_i^{\gamma\gamma}$ is the standard deviation in category i, $\sigma_{\sigma_i^{\gamma\gamma}}$ is the uncertainty on the standard deviation, $\sigma_{m_i^{\gamma\gamma}}$ is the uncertainty on the invariant mass, B_i is the number of background events, and S_i is the number of signal events. This loss function encourages the minimization of the combined relative resolution of the categories and the maximization of the signal-to-background ratio. Due to the complicated behavior of the loss function as a function of changes in the location of the boundaries, a Monte-Carlo strategy paired with a modified Nelder-Mead minimization algorithm [**?**] is used to find the boundaries corresponding to a global minimum. The Monte-Carlo approach generates random boundaries in the diphoton BDT space, and the minimization algorithm minimizes the loss function by using the random boundaries as its starting location. The process of generating and optimizing random boundaries is repeated 2000 times. Each set of boundaries and their minimized loss score is recorded, and the boundaries that produce the lowest loss are returned as the optimal boundaries.

6.3.3 Optimal number of categories

As mentioned, the optimal number of categories is determined by observing the drop-off in the gain and loss function as a function of the number of categories. The number of categories tested ranged from one to six, and the value of the loss function as a function of the number of categories is shown in Figure **??**. It was found that four analysis categories for each year's dataset was optimal.

6.3.4 Optimized boundaries and categories

Table **??** lists the final boundaries derived for each dataset. To further improve the resolution, and to minimize the systematic uncertainties affecting the mass measurement, the categories are broken down by kinematic features. The "cleanest" events are those where both photons are in the EB ("BothEBUntaggedTag_ X), and the remaining events are grouped into a single "NotEBEBUntaggedTag_ 0" category, i.e., the highest diphoton MVA category. To further reduce the systematic uncertainty on the mass measurement, events with photons in the forward region of the EE ($|\eta| > 2.1$) are dropped. This restriction is discussed in Chapter **??**.

Chapter 7

Statistical analysis

7.1 Introduction

The CMS Collaboration uses Bayesian statistics to measure values in data and estimate their associated uncertainties. This chapter outlines the method of constructing a test statistic, the treatment of nuisance parameters and systematic uncertainties, and the process of building signal and background models.

The signal and background yields are subject to systematic uncertainties, introduced as nuisance parameters (θ) such that the signal and background yields are functions of θ: $s(\theta)$ and $b(\theta)$. Uncertainties are assumed to be either fully correlated (100%) or uncorrelated (0%), and uncertainties that are partially correlated are broken down into components that are chosen to be 100% or 0% correlated. The choice of correlation in these cases is based on what is deemed most appropriate or more conservative.

Following the Bayesian approach to uncertainties, probability density functions (PDFs) are constructed for the systematic uncertainties $\rho(\theta|\tilde{\theta})$, where $\tilde{\theta}$ is the default value of the nuisance parameter as determined by external studies. These PDFs reflect the degree of belief in the true value of θ, but evaluating this is challenging both philosophically and in practice. Instead, it is helpful to perform a "change of view" on the PDFs to represent them in a frequentist context. To do this, we apply Bayes' theorem:

$$\rho(\theta|\tilde{\theta}) \sim p(\tilde{\theta}|\theta)\pi_\theta(\theta), \tag{7.1}$$

where $\pi_\theta(\theta)$ is a set of functions describing hyper-priors for those measurements. This change of view is easily incorporated into our test statistics, and $\pi_\theta(\theta)$ can be kept flat. By introducing hyper-priors and changing to a frequentist interpretation of our systematic uncertainty PDFs, the nuisance parameters are allowed to float and can be used to constrain the likelihood of the measurement [**?**].

7.2 Likelihood ratio test

Continuous physical parameters in CMS are measured using a maximum likelihood estimation. This process relies on the likelihood function ($\mathcal{L}(\text{data}|y)$), where data depends on the parameter of interest (POI) y. In the case of this analysis, data is represented by the measured values of $m_{\gamma\gamma}$, while y represents the value of m_H that maximizes the likelihood function.

The H$\rightarrow \gamma\gamma$ analysis is an unbinned analysis that uses PDFs of observables (x), built from simulation and data sidebands (described later), for signal ($f_s(x|\theta)$) and background ($f_b(x|\theta)$). These PDFs are weighted by signal and background event weights: $S(\theta)$ and $B(\theta)$. The signal PDF and event rate are also scaled by a factor that depends on the cross-section and the branching ratio expected for the SM Higgs boson, known as the signal strength μ. The likelihood function for a parameter of interest y is then:

$$\mathcal{L}(\text{data}, \tilde{\theta}|y, \theta) = \frac{e^{-(\mu S(\theta)+B(\theta))}}{N} \prod_i^N \mu S(\theta) f_s(x_i; y, \theta) + B(\theta) f_b(x_i; y, \theta) \cdot \prod_j p_j(\tilde{\theta}_i|\theta_i), \tag{7.2}$$

where the product over i runs over all events, the product over j runs over all nuisance parameters, $\tilde{\theta}$ are the default values of the nuisance parameters (as determined from external studies), and θ are the best-fit values of the nuisance parameters. Each nuisance parameter is subject to constraints, enforced by choice of PDF for a given nuisance parameter, that penalize the fit for deviations from $\tilde{\theta}$.

7.2.1 Extracting results using the likelihood ratio test

To make determinations of continuous variables, a “best fit" value is determined by numerically minimizing $-2\ln\mathcal{L}(\text{data},\tilde{\theta}|y,\theta)$. This minimization is handled by the RooFit software package [**?**]. The best fit POI and the set of nuisance parameters that maximize the likelihood function are designated as $\hat{y}$ and $\hat{\theta}$. Using this maximum likelihood function, the negative log-likelihood function is defined as

$$-2\Delta\ln\mathrm{L} = -2\ln\frac{\mathcal{L}(\text{data},\tilde{\theta}|y,\hat{\theta}_y)}{\mathcal{L}(\text{data},\tilde{\theta}|\hat{y},\hat{\theta})} \tag{7.3}$$

where $\hat{\theta}_y$ is the set of nuisance parameters that maximize the likelihood for a given value of the parameter of interest y. To build some intuition for this test statistic, it is useful to know that small values of $-2\Delta\ln\mathrm{L}$ indicate good agreement of data and y, while large values indicate an increased incompatibility between data and y. The asymptotic limit of Equation **??** follows a χ^2_n distribution, where n is the number of dimensions expressed by the POIs y [**?**]. Confidence intervals, associated with the p-value for a given confidence level $1-p$, are derived by evaluating the regions in y that result in values of Equation **??** less than the inverse of the cumulative function of the χ^2_n distribution evaluated at 1-p: $F^{-1}_{\chi^2_n}(1-p)$. For a one-dimensional measurement, such as the mass of the Higgs boson, confidence intervals at 1 and 2 σ (68% and 95.4%) are defined by the intervals found by constraining Equation **??** to be strictly less than 1 and 4, respectively.

7.2.2 Nuisance parameters in likelihood ratio tests

In the H$\rightarrow\gamma\gamma$ analysis, all nuisance parameters affecting the signal shape are modeled with a Gaussian PDF, and those affecting the signal yields are modeled with a log-normal PDF to avoid negative signal yields. In the minimization of the likelihood function targeting a POI y ($-2\ln(\mathcal{L}(\text{data},\tilde{\theta}|y,\theta))$), the nuisance parameters θ are penalized for deviations from their default values $\tilde{\theta}$ by the choice of PDF for each nuisance parameter. This penalty constrains the values of the nuisance parameters allowing them to float without allowing them to drift

to unreasonable values. This process of constraining nuisance parameters is known as profiling. For any value of the POI y there are a set of nuisance parameters θ_y that increase the likelihood ($\mathcal{L}(\text{data}, \tilde{\theta}|y, \theta)$), or, equivalently, decrease the value of $-2\ln\mathcal{L}(\text{data}, \tilde{\theta}|y, \theta)$. Translating this feature to the test statistic $-2\Delta\ln\text{L}$, used for determining confidence intervals, results in a widening of the curve and an increase in the width of the confidence intervals, and the uncertainty on the POI y increases.

The total uncertainty contributed to the confidence interval from a set of nuisance parameters, θ, is evaluated by fixing the values of θ to their best-fit values $\hat{\theta}$ in the fit of Equation **??**. With the values of the nuisance parameters fixed, the test statistic curve $-2\Delta\ln\text{L}$ will be narrower than when they are profiled. Hence, the total uncertainty from a set of nuisance parameters θ can be extracted by evaluating the quadrature difference between the width of the curve where the nuisance parameters are profiled and where they are fixed. This process can be generalized for any subset of nuisance parameters θ', with the remaining nuisance parameters fixed to their best-fit values, while only the nuisance parameters in θ' are profiled, directly evaluating their contribution to the total uncertainty. It is important to note that the extracted uncertainties from a subset of nuisance parameters are meaningful only when the subset contains nuisance parameters that are uncorrelated or negligibly correlated with the remaining nuisance parameters.

To evaluate the contribution to the uncertainty of single nuisance parameters in the case of correlated nuisance parameters, an *impact* study is performed. The *impact* is the shift in a POI y observed by fixing a single nuisance to its $\pm1\sigma$ values and performing the fit. This method preserves the effect of correlations since the remaining nuisance parameters are profiled in two fits. The impact of a particular nuisance parameter on a POI y is a measure of the degree of correlation between them, and an impact study can be used to identify those variables that have the greatest effect on the uncertainty of y.

7.3 Signal modeling

Making statistical interpretations of observed data requires knowledge of the overall efficiency, the acceptance, and the shape of the diphoton invariant mass distribution in each category and year. Additionally, the choice of the vertex affects the shape of the $m_{\gamma\gamma}$ distribution, so the modeling is performed separately for the cases of right and wrong vertex (RV, WV) selection, defined according to the 1cm threshold. The final result is the combination of the two. As the detector performance varies from year to year, signal models are constructed separately for each year's data collection. This approach allows for variations in the diphoton mass resolution and year-dependent systematic uncertainties in the final fit.

The $m_{\gamma\gamma}$ distribution in each category is parameterized by a sum of up to five Gaussian functions. The parameterizations from simulated signal samples at seven Higgs boson "mass points" (120, 123, 124, 125, 126, 127, and 130 GeV) across the four production modes (ggH, VBF, VH, and ttH), which are combined to form a single parametric model with a simultaneous fit of all the signal samples. The individual parameters of the functional form are themselves polynomials of m_H. The number of Gaussian functions chosen in any given fit is determined by an F-test, where the "best fit" model is the one that minimizes the χ^2/n_{dof} of the fit. Here, the degrees of freedom is the number of bins in the diphoton invariant mass (which must contain at least one event) minus the number of shape parameters used in the fit. Overfitting, when observed, is mitigated by reducing the number of Gaussian functions in the fit. The parameters of each Gaussian used in the fit are determined from a fit to the $m_{\gamma\gamma}$ distribution in simulation generated with 125 GeV as the default value of the Higgs boson mass. However, during the evaluation of the confidence intervals, the value of m_H is fixed at 125.38 GeV. To account for this, the mean values of the Gaussian functions are shifted up by 0.38 GeV. This shift assumes that variations in the signal models at 125 and 125.38 GeV are minor and that any variations that do exist are covered by systematic uncertainties on the shape of the diphoton invariant mass distribution, as discussed in Section **??**.

When there are insufficient simulated events in a fit for a production mode or mass point, the signal shape is replaced by that of the ggH production mode in the Untagged 0 category

for that year, and the RV fraction is set to 1. This replacement is motivated by the fact that events sharing an experimental phase space usually have a similar diphoton mass resolution.

7.4 Background modeling

The diphoton background is modeled with analytic functions fit to the diphoton invariant mass distribution in data, in each analysis category, in the range $100 < m_{\gamma\gamma} < 180$ GeV, excluding the signal region described by $115 < m_{\gamma\gamma} < 135$ GeV.

There is no theoretical prediction of a functional form for the shape of this background, so several families of functions are tested as candidates for the analytic model. Since the measurement of the Higgs boson mass relies on a narrow and relatively small signal on a large background, the choice of function should not bias the background estimate obtained from the fit. An uncertainty associated with the choice of background function is used to account for possible changes in the background estimate.

7.4.1 Discrete profiling method

The details of the discrete profiling method are given in Ref [**?**] and are summarized here. A separate viable approach to determining the effect of a nuisance parameter on the likelihood curve is to treat the nuisance as discrete. In this case, the nuisance is fixed to a series of discrete values, and the envelope of the discrete profiles approximates the full profile. An illustration of this is shown in Figure **??**.

The discrete profiling method is used in the H$\rightarrow \gamma\gamma$ analysis to account for the uncertainty associated with the choice of the background model. As the background appears in the diphoton invariant mass spectrum as a falling exponential, it is modeled using four different families of functions: Bernstein polynomials, exponentials, Laurent series, and a power law. The form of these families is shown in Table **??**. Each family is shown to order n and is uniquely described by the coefficients a_i.

For a given family, the lowest order (n) function and the next highest order ($n+1$) function fit the background. The difference in the negative log-likelihood of the fits is

evaluated $\Delta = 2(\ln \mathcal{L}_{n+1} - \ln \mathcal{L}_n)$, and a decision is made on whether the order $n+1$ or n models the background better. Quantitatively, Δ should be distributed as a χ^2 distribution with m degrees of freedom, where m is the difference in the number of free parameters between the functions of order $n+1$ and n. For example, the exponential family of functions with order n have $2(n+1)$ free parameters, while exponentials of order $n+1$ have $2(n+2)$ free parameters, so $m = 2$. The decision of whether the function fits the background is made by evaluating the p-value using the following equation:

$$\text{p} - \text{value} = p(2\Delta NLL > 2\Delta NLL_{n+1} | \chi^2(m)). \tag{7.4}$$

The higher-order function $(n+1)$ is preferred when the p-value is less than 0.1. This function is then included in the list of functions, and the procedure continues with $n+1 \rightarrow n$. The higher-order function is deemed too flexible if the p-value exceeds 0.1, and the process finishes with n being the highest-order suitable function. This process is repeated for each family of functions.

7.5 Systematic uncertainties

This section discusses the sources of systematic uncertainty in this measurement. These uncertainties primarily affect the signal model, since the background model is derived using a data-driven technique. The uncertainties that modify the shape of the $m_{\gamma\gamma}$ distribution are built into the signal model as parametric nuisance parameters. For those that do not affect the shape of the $m_{\gamma\gamma}$ distribution the systematic variations of these parameters are treated as log-normal uncertainties. Those that do not directly affect the $m_{\gamma\gamma}$ distribution, but affect the input to the classification MVA, are treated as correlated log-normal uncertainties on the category yields to account for possible category migration.

In this section the systematic uncertainties common to all H$\rightarrow \gamma\gamma$ analyses are discussed, then the systematic uncertainties affecting the electron energy scale propagated to the photon energy are presented, and then additional systematic uncertainties affecting photon energies are described.

7.5.1 Systematic uncertainties common to all H→ $\gamma\gamma$ analyses

The following uncertainties are common to all H→ $\gamma\gamma$ analyses and will be discussed briefly:

- Theory uncertainties:
  - α_s *uncertainty*: the uncertainty on the value of the strong force coupling constant. The effect on overall yield and the effect on the categorization of events are handled separately. The variation in the relative event yield due to this uncertainty is 2.6%. Category migration effects are factorized from the yield uncertainty by dividing by the overall effect on signal normalization.
  - *parton density function (PDF) uncertainties*: the effect of the uncertainty from the choice of PDF is assessed by estimating the relative yield variation in each process and category after reweighting the events of the simulated signal sample. The effects on the overall yield and categorization are handled separately. Category migrations associated with this uncertainty are less than 1%.
  - *QCD scale uncertainty*: an uncertainty related to varying the renormalization and factorization scales. The effect of this on the uncertainty of the yield and event categorization are handled separately. For the yields, a separate scaling is implemented for each production mechanism. The migration effects, arising from three scenarios with up/down variations of the renormalization and factorization scales, are separated from the yield uncertainty by diving through the overall effect on signal normalization.
  - *Uncertainty on the H→ $\gamma\gamma$ branching fraction*: this uncertainty is estimated to be about 2% [**?**].
  - *Underlying event and parton shower uncertainty*: this uncertainty corresponds to the choice and tuning of the physics generator. The two variations are treated separately with dedicated ggH and VBF samples for each. The effect on the yields in each Untagged category is included with correlations.

- *Integrated luminosity*: this uncertainty is estimated from data and amounts to a 2.5% uncertainty on the signal yield.

- *Photon preselection*: this systematic uncertainty is the uncertainty on the ratio between the efficiency measured in data and simulation. Based on the category, it ranges from 0.1% to 0.7%, resulting in an event yield variation from 0.2% to 0.5%.

- *Photon identification BDT score*: to cover the discrepancies between data and simulation, the BDT is varied by ± 0.03, and the maximum difference is used as the uncertainty. This conservative uncertainty is propagated to the uncertainty on the signal yields in the analysis categories.

- *Per photon energy resolution estimate*: this uncertainty is parameterized as a rescaling of the resolution estimate by $\pm 5\%$ about the nominal value.

- *Trigger efficiency*: the trigger efficiency is measured from Z→ee events using the tag-and-probe technique. The size of the uncertainty varies between the photon categories, which are binned in R_9, $|\eta|$, and p_T. This uncertainty results in event yield variations of at most 0.1%.

- *Vertex finding efficiency*: this uncertainty is built into the signal model as a nuisance parameter that allows the fraction of events in the right/wrong vertex scenario to change. The uncertainty on the vertex selection efficiency is 2%.

7.5.2 Systematic uncertainties affecting the electron energy scale propagated to the photon energy

Uncertainty on the electron residual energy scale and additional smearing corrections

The residual photon energy scale and smearings are derived using electrons from Z→ee events but are applied directly to photons and are used to improve the agreement between data and simulation. As described in Section **??**, the scale in data is aligned to the scale in simulation,

and the simulation is "smeared" to match the resolution of the m_{ee} distribution in data. The agreement in data and simulation is well described by fitting the m_{ee} distributions with a Breit-Wigner function [**?**] convoluted with Crystal Ball function [**?**] and evaluating the ratio of the extracted means. The formula for this is

$$R = \frac{\mu_{ee}^{\text{data}}}{\mu_{ee}^{\text{sim}}}, \tag{7.5}$$

where μ_{ee}^{data} is the mean extracted from the fit to data, and μ_{ee}^{sim} is the mean extracted from the fit to simulation. There are several variables used in the derivation of the scales and smearings that impact their values: choice of R_9 boundary to define low-bremsstrahlung and high-bremsstrahlung objects, choice of leading electron E_T threshold in the event selection criteria, and the choice of working point (loose, medium, or tight electron IDs). A variation is imposed to determine the uncertainty associated with these variables, and the m_{ee} distributions, with scales and smearings applied, are produced for data and simulation. The m_{ee} distributions produced under a variation are fit with a Breit-Wigner convoluted with a Crystal Ball, the ratio of the extracted means is evaluated under the variation, and the uncertainty is evaluated by evaluating the ratio of the ratio of means:

$$\sigma = 1 - \frac{R_{\text{variation}}}{R_{\text{nominal}}} = 1 - \frac{\mu_{\text{variation}}^{\text{data}}}{\mu_{\text{variation}}^{\text{sim}}} \cdot \frac{\mu_{\text{nominal}}^{\text{sim}}}{\mu_{\text{nominal}}^{\text{data}}} \tag{7.6}$$

where "variation" denotes distributions made under a variation of a variable listed above, and "nominal" denotes distributions made under nominal event selections. The variations used for each variable are as follows: the low-bremsstrahlung boundary defined by a cut in R_9 at 0.96 is varied by ± 0.005, the working point is varied to the medium working point, and the tight working point and the cut on transverse momentum for the leading electron is varied by ± 2 GeV. Table **??** lists the uncertainties in quadrature calculated for eight categories.

Non-linearity

The scales and smearings are derived using Z→ee events, and the corrections are applied directly to photons from H→ $\gamma\gamma$ events. There is a possible mismatch between the energy spectrum of these two processes. We assign a systematic uncertainty to cover this effect, evaluated using the residual E_T dependence of the agreement between data and simulation. Since the energy scale corrections were derived in bins of E_T of the electron, the differences between data and simulation in these E_T bins were used for the estimate of this uncertainty source. The difference is obtained by fitting the dielectron invariant mass shape with a Breit-Wigner convoluted with a Crystal Ball function [**?**] in both data and simulation using events with two electrons in the same E_T bin. The average of these residual scale differences in percent are summarized in Table **??**.

Another useful metric for determining the uncertainty of the residual scale non-linearity is the residual difference in scale between data and MC as a function of the dielectron p_T. Events in data and simulation are binned by the transverse momentum of the dielectron parent object (p_T(Z)). The chosen binning scheme is $p_T(\mathrm{Z}) = 0, 20, 40, 60, 80, 100, 200, 400, \infty$. The m_{ee} distribution in each bin is fit with a Breit-Wigner convoluted with a Crystal Ball function to extract the central value of the distribution. The ratio of central values for data and simulation are plotted, and the systematic uncertainty is determined by eye. These plots are shown in Figures **??**, **??**, **??**, and **??**.

7.5.3 Additional systematic uncertainties affecting photon energies

The remaining uncertainties arise from the imperfect modeling of the differences between electrons and photons in simulation. These uncertainties are assumed to be fully correlated among the signal categories.

Uncertainty on the material

Any difference between the material in the tracker and its simulation will lead to a mismodeling of the difference between electrons and photons. To evaluate this effect due to

uncertainties on the material budget, Monte Carlo samples of Drell-Yan events and gluon-gluon fusion H$\rightarrow \gamma\gamma$ events with different material scenarios ($\pm$ 10%, $\pm$ 5% with respect to the nominal scenario) were simulated. We define six categories of photons (or electrons) in bins of η and R_9. The "photon scale" is defined as the ratio of the reconstructed to the generated energy of the photon. The scale distributions are fit with three Gaussians to compute the maxima of the scale for the nominal and the different material variations scenarios.

- Category 1: The photons (or electrons) are in the central barrel ($|\eta| < 1.0$) with low R_9 ($R_9 < 0.96$).
- Category 2: The photons (or electrons) are in the central barrel ($|\eta| < 1.0$) with high R_9 ($R_9 > 0.96$).
- Category 3: The photons (or electrons) are in the outer barrel ($1.0 < |\eta| < 1.5$) with low R_9 ($R_9 < 0.96$).
- Category 4: The photons (or electrons) are in the outer barrel ($1.0 < |\eta| < 1.5$) with high R_9 ($R_9 > 0.96$). We do not have enough statistical power to perform a fit in this category.
- Category 5: The photons (or electrons) are in the endcap ($|\eta| > 1.5$) with low R_9 ($R_9 < 0.96$).
- Category 6: The photons (or electrons) are in the endcap ($|\eta| > 1.5$) with high R_9 ($R_9 > 0.96$).

The total uncertainty quoted in Table **??** assumes a 5% uncertainty in the material budget.

The non-uniformity of light collection in the ECAL crystals

There is a difference in the energy scale of electrons and photons due to the non-uniformity in light collection that is caused by radiation damage to the ECAL crystals. This difference

is not accounted for in the photon energy regression because radiation damage to the ECAL crystals is not included in the simulation of the ECAL. In the previous measurement of the Higgs boson mass in the diphoton channel performed by CMS [?] this was accounted for with a systematic uncertainty, and was one of the largest sources of uncertainty at 0.11 GeV. The model used to evaluate this uncertainty indicated that the additional radiation dose from 2017 and 2018 data taking would result in even larger uncertainties, leading to a non-competitive precision on the Higgs boson mass measurement. Instead, a correction is applied to the photon energy scale to account for shower differences due to radiation, which reduces the magnitude of this uncertainty. The derivation of this correction and its uncertainties are presented in this section.

The profile of energy deposited along the length of a crystal, $\mathrm{E_{dep}}(z)$, is slightly different for photons and electrons, with the location of the shower maximum for photons about 0.85 radiation lengths, or 7 mm, further from the front face, since photons convert to an electron-positron pair before initiating their shower. Figure ?? shows the shower profiles for 100 GeV electrons and photons. This profile is well modeled by the Longo-Sestelli function [?]:

$$\frac{dE}{dz} = az^{bt_max}e^{-bz}, \tag{7.7}$$

where z is the depth, t_{max} is the location of the shower maximum, and a and b are free parameters. Figure ?? shows the extracted shower maxima for electrons and photons as a function of energy. The average difference in shower maxima is extracted from a fit and found to be 0.85 χ_0, just under one radiation length. The efficiency with which light is collected as a function of depth is called the light collection efficiency (LCE). The LCE varies as a function of depth due to the formation of color centers. Parts of the crystal that experience higher radiation develop more color centers, inhibiting light collection. The signal from the photodetectors is then:

$$\int E_{dep}(z) * LCE(z)dz,$$

and the normalized signal S is:

$$S = \frac{\int E_{dep}(z) * LCE(z)dz}{\int E_{dep}(z)dz}. \tag{7.8}$$

The transmission of light in ECAL crystals can be approximated by a light attenuation coefficient μ. Initially, when the crystals were first installed, the light attenuation length was small with typical values of $\mu_{\text{nat}} = 0.30 \pm 0.01$ m^{-1}, corresponding to an optical attenuation length of $\sim$3 m in the 22 cm crystals. At that time, the LCE was nearly uniform with slight differences due to the geometry of the crystals. After irradiation the LCE varies as a function of depth, and the loss in transparency due to irradiation is modeled by adding an induced light attenuation coefficient μ_{ind} such that the transparency is

$$T_{\text{final}} = T_0 * e^{-(\mu_{\text{nat}}+\mu_{\text{ind}})L} \tag{7.9}$$

where T_{final} and T_0 are the final and initial light transmission, and L is the average light path length through the crystal.

Since the radiation dose to the crystals increases with time, and the residual photon energy scale corrections are derived using electrons from Z boson decays, a correction is needed to compensate for the difference in signals from electrons and photons. This correction is called the 'front non-uniformity factor' (FNUF) since the non-uniformity is largest around the shower maximum for electrons, near the front face of the crystal. It is applied to photon energies as:

$$F = \frac{S_\gamma}{S_e} = \frac{\frac{\int E^\gamma_{dep}(z) * LCE(z) dz}{\int E^\gamma_{dep}(z) dz}}{\frac{\int E^e_{dep}(z) * LCE(z) dz}{\int E^e_{dep}(z) dz}}, \tag{7.10}$$

where $E^e_{dep}(z)$ and $E^\gamma_{dep}(z)$ are the energy deposition profiles for electrons and photons of the same energy.

The dependence of the LCE on radiation damage is estimated with the ECAL Longevity Model (ELM) [**?**]. This model uses a uniform instantaneous luminosity and a FLUKA [**?**] simulation to estimate the fluxes of hadrons, electrons, and photons. Values of μ_{ind} are estimated across the entire ECAL ($0 < |\eta| < 3$ in steps of 0.1) for different values of integrated luminosity. Each scenario is simulated using the S-Litrani [**?**] light simulation and accounts for the loss of gain and sensitivity in the photodetectors.

The ELM derives LCE models in bins of $|\eta|$ and integrated luminosity, and from the integrated luminosity it derives values of R/R_0. Hence, by knowing the value of R/R_0 in data, an LCE model can be selected for every event. The measured values of R/R_0 in the EB and EE, shown in Figures **??** and **??** taken from [**?**], are used to select corresponding LCE functions, which are shown in Figure **??** [**?**]. FNUF corrections, as a function of energy, $|\eta|$, and R/R_0 are computed using photon and electron showers simulated at several energies and weighted with the LCE functions, as shown in Equation **??**. These corrections are shown in Figure **??** [**?**]. Note again that this correction is only relevant to photons whose shower maximum significantly differs from that of electrons. The shower maximum of low R_9 ($R_9 < 0.96$) photons, having likely converted to an electron-positron pair before the ECAL, is nearly identical to low R_9 electrons. The shower maxima for low R_9 photons and electrons are shown in Figure **??**. The similarity of the location of the shower maxima means that no FNUF correction is needed for these photons, and no correction is made.

The agreement between the FLUKA+S-Litrani simulation and the data determines the uncertainty on the FNUF correction. The sources of this are:

- Luminosity modeling: The FLUKA simulation calculates μ_{ind} assuming a constant instantaneous luminosity. This does not include in-fill and fill-to-fill variations in the luminosity. An uncertainty of 2% in EB and 8% in EE was derived using the spread of FNUF corrections in bins of η to account for these variations in luminosity.

- Agreement between the data and S-Litrani simulation: in [**?**] the light loss of proton-irradiated crystals with $0.1\ \text{m}^{-1} < \mu_{\text{ind}} < 10\ \text{m}^{-1}$ was measured using cosmic ray muons and compared with the S-Litrani simulation. The data and simulation agreed to within 20%, and an uncertainty of 10% in EB and 20% in EE is assigned to cover "imperfections" of simulation predictions for a given value of μ_{ind}. The value is higher in the EE due to the larger radiation dose, leading to larger values of μ_{ind}.

- Changes in the of sensitivity of VPTs and APDs: The range of the effect of radiation on the EE VPT conditioning was measured to vary by up to 30% [**?**]. An uncertainty of 27% in the EE is assigned to cover these variations. The effect of radiation on the

APDs is to increase the dark current. This is tracked continuously and corrected for by increasing the applied bias voltage, so no uncertainty is needed in the EB.

- Spread in μ_{nat} measurements: The ELM model assumes an initial value of $\mu_{\text{nat}} = 0.3$ m^{-1}, however the initial measurements of μ_{nat} ranged from 0.2 - 0.35 m^{-1} [?]. An uncertainty of 15% is applied to the FNUF correction in EB and EE to account for this variation.

- Effect of "PN-corrections" on laser measurements: A drift of 1% every 10 fb^{-1} is visible in the EB energy scale in the Run 2 dataset [?]. This drift is not fully corrected by applying the laser corrections to the crystal energy calibration. The origin of this drift is not fully understood. Therefore the uncertainty associated with this effect is propagated directly to the mass measurement itself: 5 MeV for 2016 preVFP, 10 MeV for 2016 postVFP, 10 MeV for 2017, and 30 MeV for 2018.

The total systematic uncertainty on the FNUF correction is 20% in EB and 35% in EE. Additionally, a systematic uncertainty of 5 MeV for 2016 preVFP, 10 MeV for 2016 postVFP, 10 MeV for 2017, and 30 MeV for 2018 is applied directly to the mass measurement. For additional details, see [?] and [?].

Uncertainty from shower shape modeling

The uncertainty due to imperfect modeling of the shower shape for both photons and electrons in the simulation is estimated using Z→ee and H→ $\gamma\gamma$ Monte Carlo samples with and without shower shape corrections. After the shower shape correction is applied, the energy regression is re-run. The e/γ double ratio of the difference with and without the correction is taken as an estimate of the systematic uncertainty. Results are summarized in Table **??** for different photon categories in η and R_9. Due to the improved agreement of the shower shape variables between Legacy 2016 data and simulation, this uncertainty is less than in our previous measurement.

7.6 Final models

The final signal and background models with all uncertainties accounted for are presented in this section. The signal models in each of the five categories for each of the four datasets are shown in Figures **??**- **??** and their background models can be seen in Figures **??**- **??**. The combined signal and background models are shown in Figure **??**, and the expected yields in each category are shown in Table **??**

Chapter 8

Higgs boson mass measurement

The expected mass of the Higgs boson in the diphoton decay channel is measured to be

$$m_{\mathrm{H}} = 125.574 \pm 0.088(\mathrm{stat.}) \pm 0.121(\mathrm{syst.})\ \mathrm{GeV}.$$

The Higgs mass likelihood profile for the combined measurement is shown in Figure **??**. This measurement is preliminary; the CMS Collaboration will be publishing a final measurement in the coming months. The precision of the Higgs mass measurement in the diphoton channel is limited by the systematic uncertainty. Previously, the mass in the diphoton channel was measured by the CMS Collaboration using the 2016 dataset to be $m_{\mathrm{H}} = 125.78 \pm 0.18(\mathrm{stat.}) \pm 0.18(\mathrm{syst.})$ GeV [**?**], and the ATLAS Collaboration measured the mass in the diphoton channel using the full Run 2 dataset to be $m_{\mathrm{H}} = 125.17 \pm 0.11(\mathrm{stat.}) \pm 0.09(\mathrm{syst.})$ GeV [**?**]. The measurement presented in this thesis is the most precise measurement of the Higgs boson mass in the diphoton channel by the CMS Collaboration to date.

Chapter 9

Summary

A measurement of the Higgs boson mass m_{H} was performed in the diphoton decay channel using 138 fb^{-1} collected at a center-of-mass energy of 13 TeV. The precision of the measurement is now limited by the systematic uncertainties associated with the energy of photons measured by the ECAL. However, after combination with the $\mathrm{H}\to\mathrm{ZZ}\to 4\ell$ measurement, this will be the most precise measurement of the Higgs boson mass until it is measured by dedicated future experiments.

This thesis presented my work on the following:

- improving the accuracy of the measurement of the energy scale and resolution of the ECAL using electrons from Z boson decays,
- optimizing the diphoton MVA boundaries to improve the signal strength and relative resolution of the mass measurement,
- introducing new methods of determining systematic uncertainties on the electron residual energy scale and additional smearing corrections,
- and introducing new methods of determining the systematic uncertainty associated with the non-linearity between energy distributions of $\mathrm{Z}\to\mathrm{ee}$ and $\mathrm{H}\to\gamma\gamma$ events.

My colleague Alexander Ledovskoy developed the ELM framework, and worked with Badder Marzocchi to derive the FNUF corrections and their uncertainties. This work drastically reduced the systematic uncertainty on the Higgs mass measurement in the diphoton channel.

My colleagues Rajdeep Chatterjee, Badder Marzocchi, Fabio Iemmi, Junquan Tao, and Zhenxuan Zhang performed the Higgs boson mass measurement using an analysis optimized for the full Run 2 dataset.

The Higgs boson mass in the diphoton channel was measured to be

$$m_{\mathrm{H}} = 125.574 \pm 0.088(\mathrm{stat.}) \pm 0.121(\mathrm{syst.})\ \mathrm{GeV}.$$